

CHAPTER 6

WATER AND THE INTERMOLECULAR FORCES

By the end of this chapter you will have covered the following material.

Science Understanding

- Observable properties, including vapour pressure, melting point, boiling point and solubility, can be explained by considering the nature and strength of intermolecular forces within a substance ([ACSCH055](#))
- The shapes of molecules can be explained and predicted using three-dimensional representations of electrons as charge clouds and using valence shell electron pair repulsion (VSEPR) theory ([ACSCH056](#))
- The polarity of molecules can be explained and predicted using knowledge of molecular shape, understanding of symmetry, and comparison of the electronegativity of elements ([ACSCH057](#))
- The shape and polarity of molecules can be used to explain and predict the nature and strength of intermolecular forces, including dispersion forces, dipole-dipole forces and hydrogen bonding ([ACSCH058](#))
- Data from chromatography techniques (for example, thin layer, gas and high-performance liquid chromatography) can be used to determine the composition and purity of substances; the separation of the components is caused by the variation of strength of the interactions between atoms, molecules or ions in the mobile and stationary phases ([ACSCH059](#))
- Water is a key substance in a range of chemical systems because of its unique properties, including its boiling point, density in solid and liquid phases, surface tension, and ability to act as a solvent ([ACSCH061](#))
- The unique properties of water can be explained by its molecular shape and hydrogen bonding between molecules ([ACSCH062](#))
- The solubility of substances in water, including ionic and molecular substances, can be explained by the intermolecular forces between species in the substances and water molecules, and is affected by changes in temperature ([ACSCH065](#))

6.1 Water and intermolecular forces

Water has unique physical and chemical properties. For a small molecule, it has a big influence on life on Earth. Water's unique properties arise partly from the strength of the intermolecular forces between the water molecules. These forces are affected by the shape of the water molecules and are important for some practical applications.

WOW

Small but powerful

Intermolecular forces are extremely small. Yet they are used by geckos to perform amazing feats. A gecko can hang from a polished surface, supporting its entire body weight, with only one toe. Aristotle wrote about these feats 2300 years ago. Now scientists think they have determined how they can happen. There are 14 000 tiny 'hairs' called setae on every square millimetre of a gecko's footpad. Each individual seta has a diameter of 5 micrometres, thinner than human hair. These tiny setae have even smaller curved pads that create billions of interactions between the gecko's foot and the surface. This explains why a 5 cm long gecko can support the equivalent weight of a 9-year-old child.

The study of how geckos adhere to a surface has influenced the design and fabrication of bio-inspired dry and reversible adhesive surfaces.



Science Photo Library/Paul D. Stewart

Figure C6.1 ▲

The feet and toes of gecko lizards use powerful intermolecular forces to support their weight



GECKO ADHESIVE FIT FOR SPIDERMAN

Watch this video to learn about making new adhesive materials that don't feel sticky.

Forces between molecules

You need to heat water to over 3000°C to break its covalent bonds and isolate hydrogen and oxygen. When ice melts at 0°C and water vaporises at 100°C, water is merely changing state. These relatively low temperatures are evidence that covalent bonds are not breaking. The bonds between the water molecules, the **intermolecular forces**, are broken. The low melting point shows that the intermolecular forces are much weaker than the covalent bonds within the molecule.

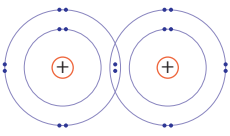
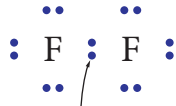
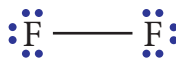
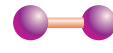
Intermolecular forces between molecules are much weaker than covalent bonds within molecules. The structure and composition of molecules determines which intermolecular forces are present. These forces determine the properties of the material.

6.2 Shapes of molecules

To explain the properties of materials, chemists need to understand their underlying structure. Molecules exist in a three-dimensional space, but we generally represent molecules in two dimensions. The structure and bonding in molecules is described by using either electron dot formulas or **valence structures**.

Electron dot formulas

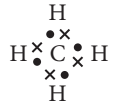
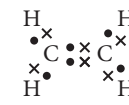
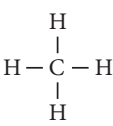
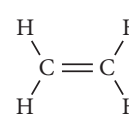
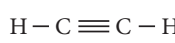
Atoms tend to be more stable when there are eight valence electrons around the central atom. This is known as the octet rule. In electron dot formulas, each of the outer valence electrons is shown as a dot (or 'x') around the chemical symbol, as shown in Figure C6.2 on page 259. Eight valence electrons would be shown by four pairs of dots circling the central nucleus. They are not shown as pairs until four positions are occupied. So, for the element nitrogen, there would be one pair and three separate electrons.

Electron valence shell diagram	Electron dot formula	Valence structure*	Physical models
 Key: $\oplus$ Nucleus • Electron	 This indicates that only one pair of electrons is shared. Key: • Valence electron F The nucleus and inner-shell electrons of fluorine	 Key: — Bonding pair F As for the electron dot formula •• Non-bonding pair	 Ball and stick  Solid ball (space-filling)
* Chemists use the convention of representing the non-bonding pairs as pairs of dots, as shown. However, some books represent them as strokes instead, or omit them from the diagram.			

▲ Figure C6.2

Representing the shape of molecules

In covalent bonds, the electrons shared between two atoms are called the **bonding pair** of electrons, so the dots are placed between the symbols for the two elements. A single bond is formed when one pair of electrons is shared. A double bond occurs when two pairs of electrons (two sets) are shared between the two atoms. The pairs of valence electrons not involved with bonding are called the **lone pair** or non-bonding pair of electrons. The lone pairs are shown around the element as a pair of dots or crosses.

Electron dot formula				 Bonding pair
Valence structure				 Lone pairs
Formula	CH ₄	H ₂ CCH ₂	HCCH	N ₂

◀ Figure C6.3

Electron dot formulas and valence structure for the molecules CH₄, H₂CCH₂, HCCH and N₂. Note that dots or 'x's are used to distinguish which electrons are from the different atoms.

Valence structures

Valence structures can be used as an alternative to drawing electron dot formulas. A single bond, shown by a line joining the two atoms, has one pair of electrons. A double bond, shown by two short lines, has two pairs of electrons. A triple bond has three short lines or three pairs of electrons. Lone pairs of electrons are drawn around the element as either two dots, a short line or another line that points away from the element. Valence structures have the advantage of showing the structure of the molecules and are often called the **structural formulas**.

Electron dot formulas use pairs of dots to represent the bonding between atoms. Valence structures use lines to show the bonding, and non-bonding or lone pairs of electrons are shown as a pair of dots, crosses or a line and can be used to represent the shape of molecules.



To revise covalent bonding and multiple bonds, refer to Chemistry section 3.5 on page 178.

ACTIVITY 6.1

REPRESENTING MOLECULES

Aim

To draw electron dot formulas and the valence structures for a range of molecules

What to do

- 1 Determine the maximum number of covalent bonds an atom of each of the following elements can form: H, C, N, O, F, Ne, P, Cl
- 2 Draw electron dot formulas for fluorine (F_2), hydrogen fluoride (HF), water (H_2O), carbon tetrachloride (CCl_4), phosphine (PH_3), carbon dioxide (CO_2), hydrochloric acid (HCl), nitrogen dioxide (NO_2) and neon (Ne).

What did you discover?

Can you use the formula to predict the arrangement of atoms around the central atom? Which atoms are more likely to be central to the molecules?

Predicting the shape of molecules

It is important to know the shape of molecules because the shape will determine the intermolecular bonding and properties of the substance. The valence structure of water shows a two-dimensional shape, in which oxygen is central to two hydrogen atoms. The three-dimensional shape could be the three atoms in a row, a V-shape molecule or two hydrogen atoms at right angles to the central oxygen.

Valence shell electron pair repulsion (VSEPR) theory has been used for over 50 years to determine the shape and hence the function of many molecules. It gives an understanding of how proteins may bond to an enzyme, essential for many medical applications.

Although the 'R' in VSEPR stands for the word 'repulsion', the theory is based upon the **Pauli exclusion principle**, not electron repulsion. Pauli's exclusion principle states that each orbital can only have 0, 1 or 2 electrons. If two electrons are present, then they spin in opposite directions around the nucleus. The discovery of this electron spin is fundamental to the Pauli exclusion principle. Two electrons of the same spin have a zero probability of being found in the same location and will be found in locations as far apart as possible.

VSEPR (valence shell electron pair repulsion) theory is based upon the number of electron pairs surrounding the central atom. The electron pairs arrange themselves as if they repel each other.

The three-dimensional structure of simple molecules depends upon the number of electron pairs around the central atom. The electron dot formulas are used to indicate electron pairs. These may be bonding and/or lone pairs. The electron pairs will be at a maximum distance from each other. It helps to think of the electron pairs as points on a sphere with the central atom in the middle.

Lone pairs are closer to the single nucleus and so occupy more space. Bonding pairs are shared by two atoms and are attracted by the two nuclei. Hence, they occupy less space and cause less repulsion than lone pairs.

The common shapes of molecules are summarised in Table C6.1 on page 262.

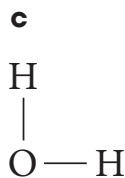
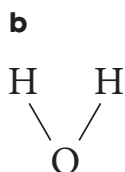
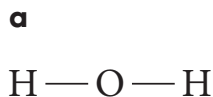
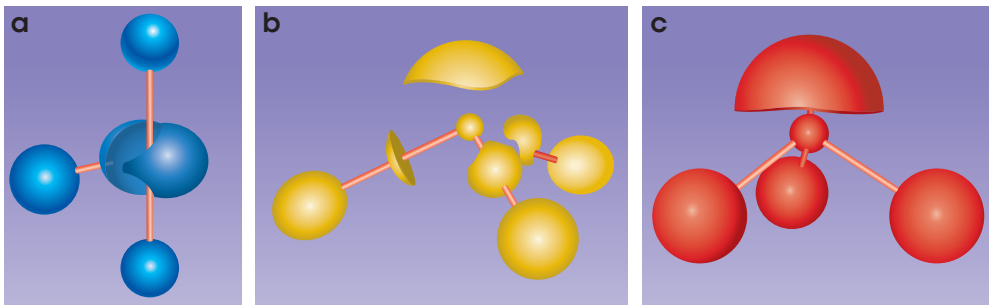


Figure C6.4 ▲

The possible shape of the water molecule: a) two-dimensional shape, b) V-shaped molecule and c) two hydrogens at right angles. Which one is correct?



▲ **Figure C6.5**

The electron density surrounding molecules calculated by the VSEPR theory. a) ClF_3 ; b) NH_3 ; c) PF_3 . Note that lone pairs take up more space than bonding pairs of electrons.

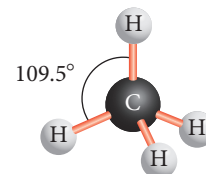
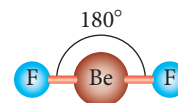
To work out the shape of a molecule:

- draw the electron dot formula
- count the lone pairs and bonding pairs. Remember that lone pairs are electron pairs that are not shared, while bonding pairs form the covalent bonds
- treat the electrons in a covalent bond, whether a single, double or triple bond, as one set of electrons
- use Table C6.1 on page 262 to predict the shape based upon the number of bonding and lone pairs.

The simplest example to show the application of the VSEPR theory is BeF_2 (Figure C6.6). Here the central Be atom has only two electron pairs, one to each F atom. The three atoms are in a line, which means that the bonding pairs are at 180° from each other. The shape is linear.

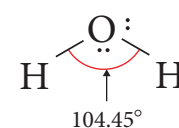
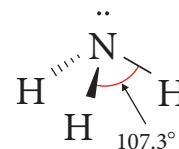
The carbon atom in methane (CH_4) has four bonding pairs to the hydrogen atoms. The four hydrogen atoms and their bonding pairs of electrons are furthest apart in a tetrahedral shape (the angle between the bonds is 109.5°).

The nitrogen atom in ammonia (NH_3) has four electron pairs, which includes three bonding and one lone pair. The hydrogen atoms are at three corners of the tetrahedral and the lone pair occupies the final spot. This results in a pyramid shape for ammonia (the angle between the bonds is 107.3°) (see Figures C6.7 and C6.8).



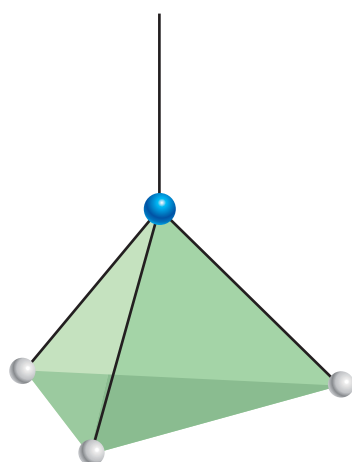
▲ **Figure C6.6**

The molecule BeF_2 is linear and CH_4 is tetrahedral.



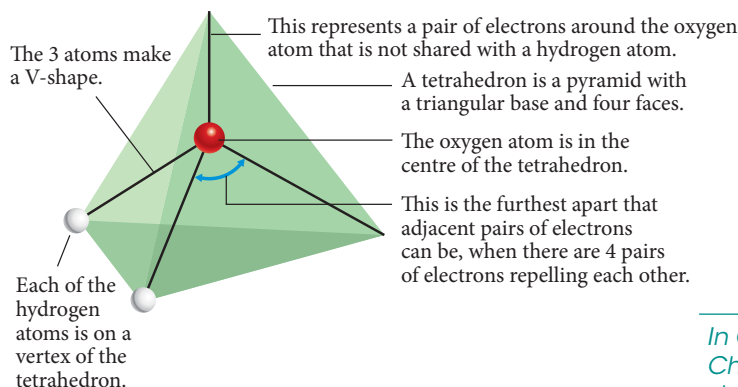
▲ **Figure C6.7**

Ammonia is pyramid-shaped and water is bent or V-shaped.



▲ **Figure C6.8**

Explaining the pyramid shape of ammonia



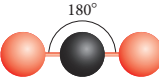
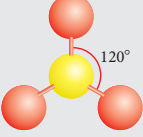
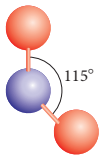
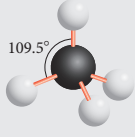
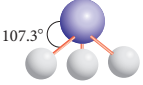
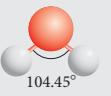
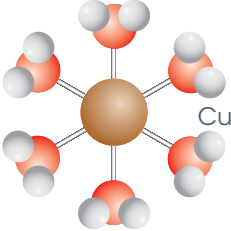
▲ **Figure C6.9**

Explaining the V-shape of water

Oxygen is the central atom in water. Oxygen has two lone pairs and two bonding pairs of electrons. These four pairs are directed to the corners of the tetrahedron. Two of these are taken up by oxygen's lone pairs. This results in a V-shape for water molecules. Water is often called a bent molecule (see Figure C6.9).

In Chemistry Chapter 3, the structure of pure carbon in diamond form was described as tetrahedral, with C atoms at each vertex of the tetrahedron. The electronic configuration of a carbon atom is $1s^2 2s^2 2p^2$.

Table C6.1 Shapes of molecules

Generic formula (M, X = atoms; E = lone pair of electrons)	Number of electron pairs	Lone pairs on M	Molecular shape
MX	1	0	Linear  HCl
MX ₂	2	0	Linear  CO ₂
MX ₃	3	0	Trigonal planar  SO ₃
MX ₂ E	3	1	V-shaped  NO ₂
MX ₄	4	0	Tetrahedral  CH ₄
MX ₃ E	4	1	Trigonal pyramidal  NH ₃
MX ₂ E ₂	4	2	V-shaped  H ₂ O
MX ₆	6	0	Octahedral  Cu(H ₂ O) ₆ ³⁺



VSEPR AND THE 3D STRUCTURES

Visit this website to see the VSEPR structures of some common molecular geometries.

ACTIVITY 6.2

MOLECULAR MODELS

Three-dimensional molecules are best viewed with the aid of models.

Aim

To produce a series of models that demonstrates the shape of molecules

You will need

- Molecular model kits or straws and beads
- Digital camera (optional)

What to do

- 1 Create a table with column headings HF, CO₂, H₂O, CH₄, C₂H₆, C₂H₄, NH₃, NO₂.
- 2 **a** For each compound, draw the electron dot formulas.
b Note the number of bonding and lone pairs.
- 3 **a** Draw the valence structure.
b Note the number of lone pairs and bonding pairs.
- 4 Use the table to predict the shape of the molecules.
- 5 Make the model of the molecule.

What did you discover?

Compare your model with your prediction. How did it compare? Was it easier to see in three dimensions?

Extension

Use the Internet, computer programs or tablet apps to find 'ball and stick' and space-filling representations of the molecules. Why would chemists have so many ways to represent the molecules?

QUESTION SET 6.1

Remembering

- 1 What is the maximum number of covalent bonds the following atoms can form?
a Cl
b O
c N
d C
e H
f S
- 2 Why does oxygen always have two lone pairs of electrons?
- 3 Draw the electron dot formulas for HCl, NH₃ and CH₄ and answer the following.
a Identify how many bonding pairs and non-bonding pairs there are in each molecule.
b Describe the shape of each of these molecules.
c Explain why they have this shape.

Understanding

- 4 Explain why H₂S has the same shape as H₂O.
- 5 Give explanations for why a water molecule is V-shaped but carbon dioxide is linear.

Applying

- 6 Draw the electron dot and valence structures and then describe the shape you would expect for each of the following molecules.
- SiH_4
 - CH_3Cl
 - H_2O_2
 - CS_2
 - H_2Se
 - HCN
 - COCl_2
 - CO
- 7 Phosphine (PH_3) is a toxic gas that consists of only P and H.
- Predict the shape of phosphine by:
 - writing the electronic configuration of P.
 - drawing the electron dot formula of P and hence deduce the formula for phosphine.
 - determining the number of bonding and non-bonding pairs.
 - predicting the shape of the phosphine.
 - Why would this molecule have a different shape from PF_5 ?

6.3 Uneven electron sharing



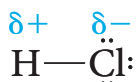
To revise electronegativity, refer to Chemistry section 1.4 on page 111.

In Chemistry 3, you considered molecules such as N_2 or Cl_2 made from the same non-metal element. In these molecules, the electrons in the covalent bonds are evenly shared between the two atoms. This is not true with certain **heteroatomic** molecules that have more than one type of element present. With molecules such as HCl , NH_3 and H_2O , the electrons are not evenly shared. That is because the three elements Cl, N and O tend to attract the electrons more, due to their **electronegativity** (electron-attracting power).

Bonding electrons in HCl molecules are strongly attracted to the chlorine nucleus because chlorine is more electronegative than hydrogen. This means the electrons will spend more time at the chlorine end of the molecule and this end of the molecule will be slightly more negative. This is represented in diagrams with a delta negative sign (δ^-). The hydrogen end will be slightly more positive and is represented by a delta positive sign (δ^+). This is shown in Figure C6.10.

The hydrogen–chlorine bond is a **polar covalent bond**. The molecule is considered to be a **dipole**: it has two poles of opposite charges at either end, rather like a magnet with a north and a south pole. Diatomic molecules with a net dipole are considered to be **polar**. Non-polar molecules such as hydrogen have a uniform charge over the whole molecule.

Common polar bonds are those between $\text{O}-\text{H}$, $\text{N}-\text{H}$, $\text{C}-\text{O}$ and $\text{C}-\text{Cl}$. The most electronegative elements in order of decreasing strength are:



▲ **Figure C6.10**
The polarity of the HCl molecule

Uneven sharing of bonding electrons can occur when H or C forms covalent bonds with strong electronegative elements such as O, N, Cl and F. When this happens, the covalent bond is a polar covalent bond.

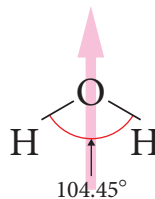
Polar and non-polar polyatomic molecules

For molecules with more than two different atoms, the presence of a polar covalent bond does not necessarily mean the molecule is polar.

Carbon dioxide (CO_2) is a linear molecule with a central carbon atom and oxygen on each side. The molecule is symmetrical. The C—O bond is polar, with the dipole pointing towards the oxygen because it is more electronegative, but there is no net separation of charge in the overall molecule. Each oxygen will attract the electrons, but as they are on either end of the molecule, the charge is distributed over the whole molecule. It is like having an equally strong person on each end of a rope in tug of war; the rope doesn't move as the pull from each end is equal. The poles cancel each other out. Although the molecule has polar covalent bonds, there is no net separation of the charges so no net dipole. This molecule is non-polar.

Contrast this with water as shown in Figure C6.11. In water, the O—H bond is polar. Oxygen is central and hydrogen is at the bottom of the Λ -shape. This means that the bottom end of the molecule is slightly positive. Water is asymmetrical; there is a net dipole present so water is polar. The presence of polar covalent OH bonds and a V-shape means that the water molecule is polar.

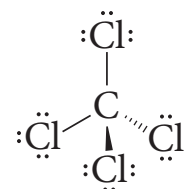
The ammonia (NH_3) and tetrachloromethane (CCl_4) molecules have different shapes and polar bonds. NH_3 , a pyramidal-shaped molecule, is polar. The N atom at the top of the pyramid is slightly negative due to its greater electronegativity. The hydrogen end of the molecule is slightly positive. Due to the asymmetry, it is a polar molecule. CCl_4 has polar bonds due to differences in electronegativity between C and Cl. It is tetrahedral in shape with a central carbon. The dipole in each of the C—Cl bonds points to the Cl atom. The negative regions on chlorine in CCl_4 cancel themselves out due to the symmetry of the molecule. CCl_4 molecules are non-polar.



◀ Figure C6.11

The carbon dioxide molecule is non-polar, despite the presence of polar covalent bonds. The water molecule is polar. The dipoles add together in water but cancel out in carbon dioxide. The arrow points from the positive end to the negative end of the dipole.

To determine symmetry, it may help to consider the symmetry of your body. Vertically you are symmetrical, with an arm and leg on either side. But horizontally above and below your waist you are not symmetrical (asymmetrical).



Molecules can be non-polar despite the presence of polar bonds. For the molecule to be polar, there must be a polar bond present and there must be asymmetry in the molecular shape.

▲ Figure C6.12

The molecular shape of ammonia (NH_3) and tetrachloromethane (CCl_4)

QUESTION SET 6.2

Remembering

- 1 List the four elements that form polar covalent bonds with carbon or hydrogen.
- 2 Define the following terms.
 - a Dipole
 - b Polar covalent bond
 - c Polar

Understanding

- 3 The most electronegative elements in order of decreasing strength are $\text{F} > \text{O} > \text{N} = \text{Cl} > \text{Br} > \text{C} = \text{S} = \text{I} > \text{P} = \text{H} > \text{Si}$. Consider the following pairs of elements: O—H, Cl—H, C—H, P—H, N—O, N—Cl, N—H, S—H, C—S, F—O, Cl—O, C—O, P—O, C—Cl.
 - a In covalent bonds between these elements, indicate which element would be slightly more negative and which would be more positive by labelling with δ^- and δ^+ where appropriate.
 - b Classify the pairs of elements given above as highly polar, having some polarity or non-polar.



Applying

4 Consider the following molecules.

- a SiH_4
- b CH_3Cl
- c CS_2
- d PH_3

- i Draw electron dot formulas and valence structures.
- ii Determine the shape of the molecule.
- iii Predict if the molecule has polar bonds.
- iv Show the direction of the dipole, if present, by drawing an arrow towards the negative end of each polar bond.
- v State whether the molecule has a net dipole, making it polar.

5 Determine whether the following are polar.

- a Ammonia
- b Iodine chloride
- c Dichloromethane (CH_2Cl_2)
- d Nitrogen trichloride
- e Sulfur difluoride

Analysing

6 Hydrogen peroxide is polar. Predict the shape of H_2O_2 . Explain why at first this appears unusual.

6.4 Three types of intermolecular forces

All bonding is due to attraction of opposites (**electromagnetic attraction**). There are three types of intermolecular forces – **dispersion forces**, **dipole–dipole forces** and **hydrogen bonds**. Intermolecular forces occur between the molecules. They can occur between molecules of the same substance and between molecules of different substances.

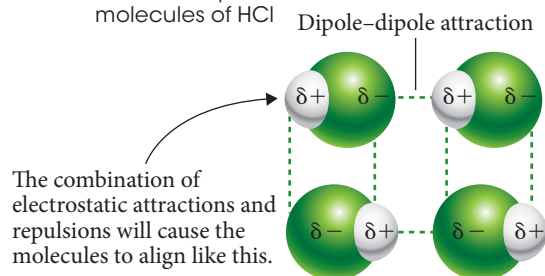
Intermolecular forces are much weaker than the intramolecular forces that exist within molecules. Intramolecular forces are generally considered to be covalent. Covalent, ionic and metallic bonding are much stronger than the intermolecular forces.

The strength of intermolecular forces increases with increasing polarisation of molecules. The weakest is the dispersion force, then the dipole–dipole force and then a special case of the dipole–dipole force, the hydrogen bond.

Dipole–dipole forces

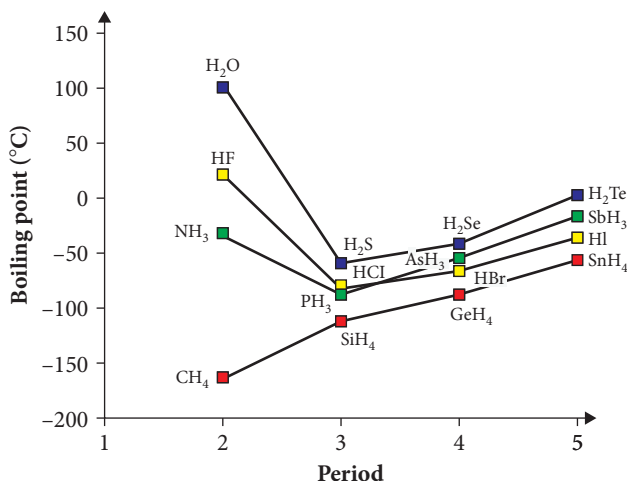
In polar molecules, there is a net dipole. One end of the molecule is slightly negative while the other end is slightly positive. Hydrochloric acid (HCl) is a polar molecule. The presence of dipoles creates an attraction between neighbouring molecules, a dipole–dipole force, which is an attraction between two dipoles in two individual molecules. Molecules line up so that the slightly positive ($\delta+$) end of one molecule attracts the slightly negative end ($\delta-$) of another molecule. The dipole–dipole force (dipole–dipole attraction) is relatively strong because the dipole is permanent.

Figure C6.13 ▼
Dipole–dipole attraction between polar molecules of HCl



Hydrogen bonds

Boiling and melting points give an estimation of the strength of the intermolecular forces between molecules. Figure C6.14 shows a general increase in boiling points as you go down the periodic table groups, if you could exclude period 2 hydrides.



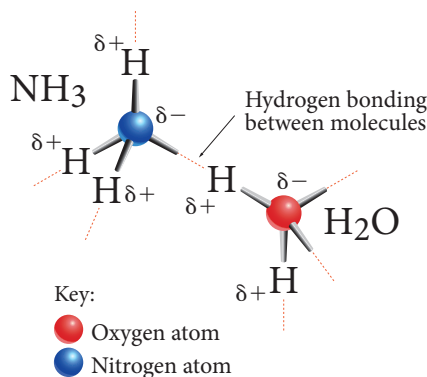
◀ **Figure C6.14**
Boiling point of hydrides of groups 14–17

When nitrogen, oxygen or fluorine is attached to a hydrogen atom, the resulting molecule has an unusually high boiling point. When these very electronegative elements bond to hydrogen, a special type of dipole–dipole force occurs called a hydrogen bond. This is the strongest of the weak intermolecular forces being about one-tenth the strength of a covalent bond. In diagrams, it is shown as a dotted line between the molecules.

Hydrogen bonds are due to the hydrogen nucleus (a single proton) being extremely small. When hydrogen is bonded to the most electronegative elements (fluorine, oxygen and nitrogen), the charge over the polar covalent bond is at a maximum. Water displays hydrogen bonding. Oxygen's strong ability to attract electrons means that the electrons in the O–H covalent bond are more attracted to oxygen. Oxygen becomes slightly negatively charged and hydrogen is left with a slightly positive charge. The hydrogen atom is so small that it also means that two molecules can get very close to each other. Similarly, in ammonia the nitrogen attracts the electrons of the covalent bonds, leaving the hydrogen atoms more positive. These can attract adjacent nitrogens. Hydrogen bonding then occurs between the hydrogen in –OH, –NH or –FH and the lone pairs on nitrogen, oxygen or fluorine on the adjacent molecule.

The dipole–dipole force occurs when there is a permanent dipole present. The hydrogen bond is a special case of the dipole–dipole force.

A hydrogen bond occurs when an H atom that is bonded to an O, N or F atom in one molecule becomes attracted to the lone pair of electrons of N, O or F of an adjacent molecule.



◀ **Figure C6.15**
Hydrogen bonding occurs between an H on one molecule and the lone pair of electrons on N (or O or F) in another molecule.

Shocking news: Hydrogen bond about to be redefined?

The definition of a hydrogen bond may be about to change. It is not as clear cut or simple as it appears. Scientists have found that hydrogen bonding is occurring in more subtle situations. Solid hydrogen sulfide appears to have a structure resembling ice. Some researchers have discovered evidence that hydrogen sulfide can form hydrogen bonds with ethylene. Even more unusual is that non-polar substances such as methane and krypton have been found to bond with hydrogen. But maybe the most controversial is the evidence of shared electron density between hydrogen-bonded molecules. This means that there is some covalent nature to this bond.



To see how different types of intermolecular forces relate to the properties of substances, refer to Context 3, 'Water: the vital substance', page 59.

Figure C6.17 ▼

Noble gases can explain how dispersion forces occur through instantaneous dipoles induced due to the movement of electrons.

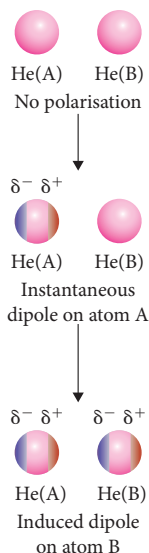
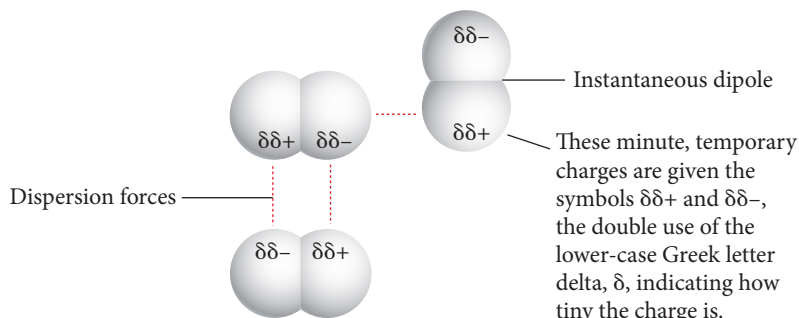


Figure C6.16 ►

The dispersion forces between hydrogen molecules in solid hydrogen, showing instantaneous dipole-dipole interactions



Dispersion forces

Polar molecules can form dipole–dipole or hydrogen bonds between the molecules. Non-polar molecules cannot, so there must be some other force that enables non-polar molecules, such as dry ice, liquid nitrogen and oils and fats, to form liquids and solids. This force is a dispersion force.

The simplest molecule, hydrogen (H_2), can be used to explain this force. The force is extremely weak because hydrogen has an extremely low melting point. The hydrogen molecules have a cloud of electron density surrounding them. When the electrons are on one side of the molecule there is, for an instant, a temporary dipole. One end of the molecule will be very slightly more positive and the other end will be very slightly more negative. The temporary positive end attracts a neighbouring molecule's electrons, inducing them to come towards the positive area. This forms the weak electrostatic attractions that are dispersion forces. The fluctuating changes in charge result in a very weak force between molecules.

Dispersion forces must exist in every molecule, as every molecule has the rapidly fluctuating electrons orbiting the molecule. The force is due to the rapid movement of electrons in the cloud of electron density surrounding the molecule.

Dispersion forces are much weaker than other intermolecular forces. When dipole–dipole or hydrogen bonding is present, the effect of dispersion forces is discounted.

Dispersion forces are also present in the noble gases (see Figure C6.17). Electrons in two adjacent atoms occupy positions so that the atoms form temporary dipoles. With more electrons, there are more chances for dispersion forces to occur. Molecules are electrically neutral, so when there are more electrons, there are more protons. So, increased molecular weight means an increased chance of dispersion forces and a higher melting point. The electrons in larger atoms are also further from the nucleus and more readily form temporary dipoles. This explains why the larger noble gas argon has a boiling point of -186°C , much higher than that of helium -272°C . A similar trend of increasing dispersion forces with increasing molecular weight is seen with the halogens.

Dispersion forces are due to instantaneous temporary dipoles that occur with the movement of electrons. The more electrons present (the larger the molecular weight), the bigger the dispersion force will be between the molecules.

A gecko's amazing ability of their toes to stick to almost anything relies on dispersion forces between each seta and the surface. With a large surface area for dispersion forces to occur, the overall force can be strong.

Table C6.2 Boiling points of noble gases and halogens

Noble gas	Boiling point (°C)	Halogen	Boiling point (°C)
Helium	−269	Fluorine	−188
Neon	−246	Chlorine	−34.6
Argon	−186	Bromine	58.8
Krypton	−152	Iodine	183
Xenon	−108	Astatine	337
Radon	−62		

6.5 The importance of bonding

Bonding is responsible for many of the properties of materials and biomolecules. The nucleic acids in the DNA double helix are held together by hydrogen bonds. This allows them to unzip easily, enabling replication and transcription. Hydrogen bonding helps maintain the structures of proteins, enzymes, ribosomes and cell membranes.

Analytical techniques such as chromatography work because of surface interactions and the intermolecular bonds of different substances to different surfaces or solvents. Utilising this knowledge allows chemists to determine the organic substances in mixtures.



INTERMOLECULAR FORCES AND BOILING POINTS

Visit this website to learn how intermolecular forces affect boiling points.



To learn more about in the interaction between molecules, refer to Chemistry section 8.2 on page 346.

Water and hydrogen bonding

There are three types of bonding in water. Strong covalent bonding exists between the H and O within the molecule. Hydrogen bonding and dispersion forces occur between the water molecules. Hydrogen bonding is responsible for the unique properties of water.

The unique properties of water include unusual melting and boiling points, density in solid and liquid phases, **surface tension** and an ability to act as a very good solvent. Hydrogen bonding explains all the properties that relate to temperature. You need a relatively large amount of energy to break the bonds in water, which is why water has a relatively high boiling point, 100°C. For a certain amount of energy, fewer molecules escape into the air. The fewer vaporised water molecules means lower **vapour pressure**, the downwards pressure exerted by the gas molecules on the surface of the liquid. To change state from liquid to gas (high latent heat) or to increase the average temperature of the water (high specific heat) energy must be absorbed. The change is endothermic.

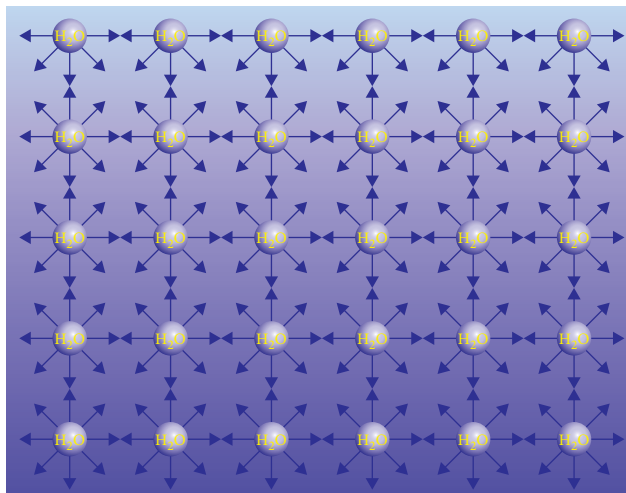
The surface tension of water is high as hydrogen bonding occurs in all directions and is relatively strong. It holds the molecules together.

The cohesive nature of water helps plants take up nutrients from the soil. Root hairs draw up water containing dissolved nutrients such as nitrogen, phosphorus and potassium from the soil. **Capillary action** helps the water to move up the xylem of the plant to the leaf surface. Capillary action relies on the adhesion of water molecules to the sides of the xylem vessel and the cohesive properties of water. The cohesion of water molecules means dissolved nutrients are pulled all the way up tall trees, defying gravity.

There is a negative correlation between intermolecular force strength and vapour pressure. Compounds that have strong intermolecular forces will have low vapour pressure.

Figure C6.18 ►

In the bulk of liquid water, each molecule has forces operating in all directions. However, on the surface, there is an imbalance – the liquid molecules are attracted to each other and exert a net force that pulls them together.



To see how hydrogen bonding affects water, refer to Context 3, 'Water: the vital substance', page 67.

Capillary action is due to three main forces. These are the:

- **cohesive force** – the intermolecular force between molecules in a substance that helps to maintain a certain shape of the liquid.
- **surface tension** – this is due to the cohesive forces at the surface of material and results in the surface of the fluid being under tension
- **adhesive force** – these are the forces of attraction between unlike molecules.

Capillary action can be seen when part of a towel is dipped in water – the whole towel eventually gets soaked. Capillary action can also be seen in narrow glass vessels, where adhesive forces between water molecules and the surface of the glass and cohesive forces in water cause the surface of the water to curve to form a **meniscus**. The water continues to be drawn up, until gravity overcomes the adhesive force.

INVESTIGATION 6.1

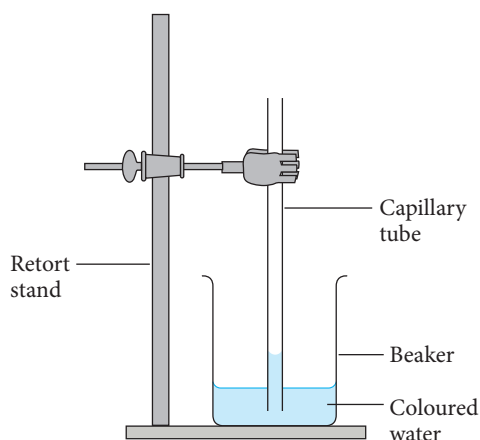
CAPILLARY ACTION

Aim

To determine the relationship between the diameter of a tube and the height of the water column

What do you need

- Beaker with coloured water
- Glass tubes with diameters in the range 1–4 mm



◀ **Figure C6.19**

Experimental set-up for testing capillary action

What to do

- 1 Complete the following risk assessment to identify any risks that may occur during the investigation and ways to reduce them. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?

- 2 Place the glass tubes upright in the beaker of coloured water. The water should be drawn up the surface of the tubes.
- 3 Measure the height of the water column.

What did you discover?

- 1 Was there a correlation between the size of the capillary tube and the height of the water column?
- 2 Did you always notice the water curving up to form a meniscus? Was the surface tension always strong enough to support a column of water defying gravity?

Extension

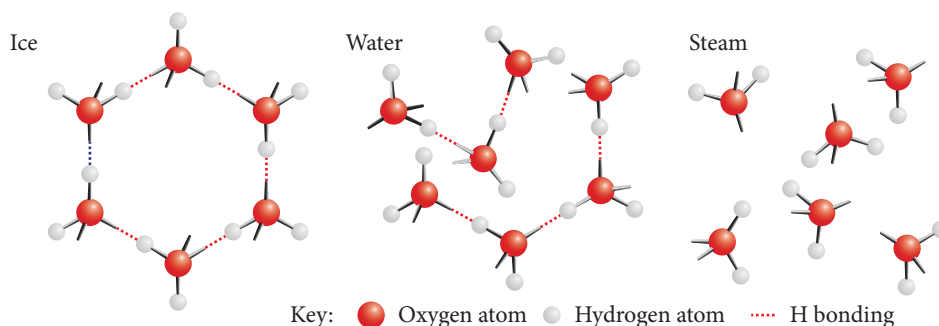
Use the Internet and other resources to investigate why mercury has a concave meniscus. How do solutes in water affect capillary action?

As with all substances, as water cools, its particles slow down and pack closer together so the density increases. Therefore, cooler water has a higher density than warmer water. When the temperature is less than 4°C, the molecules do not have as much kinetic energy and no longer move between the spaces, but form more hydrogen bonds, as shown in Figure C6.20. The shape of the water molecule means a regular hexagonal shape is formed. Water has the unique property of expanding as it freezes. The density decreases as it cools from 3°C to 0°C. Ice is about 8% less dense than water.

Another property of water is its ability to act as a solvent. Pure water is liquid from 0°C to 100°C. This means that it can act as a solvent over this temperature range. Water dissolves and transports a range of material across the whole planet and also through each cell of a living organism. Water does this by forming hydrogen bonds or **ion-dipole** bonds with a wide range of substances from salts to organic molecules such as proteins.



To revise how the density of water and ice influences the physical and biological processes on Earth, refer to Context 3, 'Water, the vital substance', page 63.



◀ **Figure C6.20**

The structure of ice compared to that of liquid water and steam showing hydrogen bonding



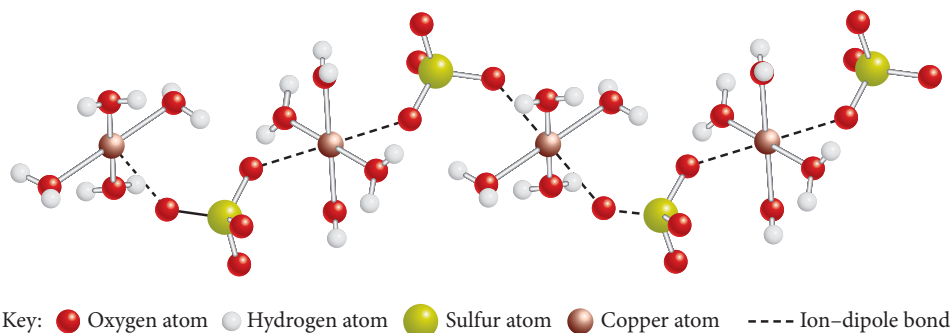
An understanding of hydrogen bonding, dipole-dipole and dispersion forces is important because it allows chemists to predict properties such as melting and boiling points and whether one substance will dissolve in another substance.

Water of crystallisation

Water of crystallisation occurs when water molecules are attracted to the ions of a salt. Many ionic compounds that crystallise out of an aqueous solution include water molecules in a regular array through the crystal lattice. The water molecules are in set ratio to the ionic substance. These compounds are called hydrated. For example, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has five water molecules of crystallisation per unit of copper(II) sulfate. The bonds between the ions and water are called ion-dipole bonds; the ion and the dipole in the water molecules form a weak bond. Water or other molecules that form dipole bonds to a metal atoms are referred to as **ligands**. The water ligands take the positions around the central ion and will follow the VSEPR theory. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is an octahedral shape. The central ion is a cation, so the negative oxygen of water points towards the Cu^{2+} as shown in Figure C6.21.

Figure C6.21 ►

The crystals of copper(II) sulfate consist of an octahedral arrangement of water molecules and sulfate ions around the central copper ion.



The water of crystallisation can be evaporated by heating the hydrated compound. When all the molecules have been removed, the compound is said to be **anhydrous**. To do this, a known mass of the hydrated compound is heated so that the mass of water removed can be determined. From this, the empirical formula of the hydrate can be calculated.

EXPERIMENT 6.1

WATER OF CRYSTALLISATION

Aim

To determine the number of water molecules of crystallisation of hydrated sample

Materials

- Hydrated samples (2–3 g) of sodium carbonate, barium chloride, calcium chloride, copper(II) sulfate, magnesium sulfate
- Crucible and lid
- Desiccator
- Matches
- Balance
- Bunsen burner
- Pipe clay triangle
- Tripod
- Crucible tongs
- Heatproof mat

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
A hot crucible retains heat, especially if its contents have significant mass.	Avoid inhaling fumes when heating. Use tongs to handle the crucible; place the hot crucible on the heatproof mat. Do not touch the hot crucible. If you burn yourself, place the affected part under cold running water for 10 minutes and inform your teacher.
The Bunsen burner will get hot.	Do not use the Bunsen burner if the gas tube is damaged. Ensure long hair is tied back and the flame is away from flammable material. If you burn yourself, place the affected part under cold running water for 10 minutes and inform your teacher.
Barium salts are moderately toxic.	Avoid breathing fine dust. Wear safety glasses and wash your hands after the experiment. Dispose of all chemicals according to your teacher's directions. Do not pour them down the sink.
Copper(II) sulfate is slightly toxic	Wear safety glasses and wash your hands after the experiment. Dispose of all chemicals according to your teacher's directions. Do not pour them down the sink.
Ammonium chloride is slightly toxic.	Use in a well-ventilated area. Wear safety glasses and wash your hands after experiment. Dispose of all chemicals according to your teacher's directions. Do not pour them down the sink.
Sodium carbonate may be a slight skin irritant.	Wear safety glasses and wash your hands after the experiment.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

- 1 Heat the clean, dry crucible and lid strongly for 5 minutes over a Bunsen flame. Place the crucible in the desiccator and allow it to cool.
- 2 Weigh the cooled crucible and lid on an accurate balance. Add 2–3 g of your hydrated crystals and immediately reweigh the crucible.
- 3 Place the lid on the crucible, leaving a small gap, and heat it for 5 minutes in a low flame. Then heat it strongly for 20 minutes. Allow the crucible to cool in the desiccator and then reweigh it.
- 4 Heat the crucible strongly for a further 5 minutes, cool it in the desiccator and reweigh it. Repeat until constant weight has been obtained; that is, to within 1–2 mg.

Results

Record your results in a table like the one below.

Mass of crucible and lid	
Mass of crucible, lid and hydrated salt	
Mass after heating of crucible, lid and salt	

Analysis of results

- 1 Calculate the mass of hydrated compound used (e.g. $\text{MgSO}_4 \cdot n\text{H}_2\text{O}$).
- 2 Calculate the mass of anhydrous compound remaining after heating, and the mass of water removed.
- 3 Refer to a table of relative atomic masses to calculate the molar mass of compound (e.g. MgSO_4 and water H_2O).
- 4 Calculate the amount of substance of water removed (i.e. $n(\text{H}_2\text{O})$).
- 5 Calculate $n(\text{sample})$; for example, $n(\text{MgSO}_4)$, remaining after heating.
- 6 Determine the ratio of salt to water: $n(\text{MgSO}_4):n(\text{H}_2\text{O})$ as a simple whole number ratio. Hence, write the empirical formula of hydrated compound in the form $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ and state the degree of hydration of the sample.

Discussion

- 1 What are the main sources of error in your experiment?
- 2 If your class analysed different compounds, collate the results for the whole class to determine the water of crystallisation for each compound.
- 4 Compare the experimentally derived formulas with the known values.
- 5 Which experimentally derived formulas were further from the known values?
- 6 Brainstorm reasons why there was a range of discrepancies; for example, did this correlate with the main sources of errors? Did the discrepancy correlate with the number of moles of samples each person had?

QUESTION SET 6.3

Remembering

- 1 Name the three:
 - a strong bonding forces.
 - b weak intermolecular forces.
- 2 Recall what causes dipole–dipole interactions and dispersion forces.
- 3 What are the common groups that exhibit hydrogen bonds?

Understanding

- 4 Explain why you should not put full water bottles in the freezer.
- 5 Rank the three intermolecular forces from weakest to strongest. Explain how polarity influences this order.
- 6 What are the two factors influencing a molecule's polar character?

Applying

- 7 If water did not exhibit hydrogen bonding, at what temperature should water boil?
- 8 Explain why H_2S , a heavier molecule with more electrons, has a lower boiling point than water.

Analysing

- 9 Water has the second highest specific heat capacity of all known substances. It takes a lot of energy to heat 1 gram of water by 1 degree. This is due to the strength of the intermolecular forces within water. Liquid ammonia has the highest specific heat. Explain why ammonia has a higher specific heat capacity than water.
- 10 Oxygen levels in water are higher near the surface. Oxygen is more soluble in colder water. These two facts do not contradict each other. Why?

Reflecting

- 11 Geckos have utilised dispersion forces to allow them to support their full weight with just one toe. If scientists could replicate the extraordinary ability of the gecko to attach and detach to smooth surfaces, explain how that knowledge could benefit our society.

6.6 Solutes

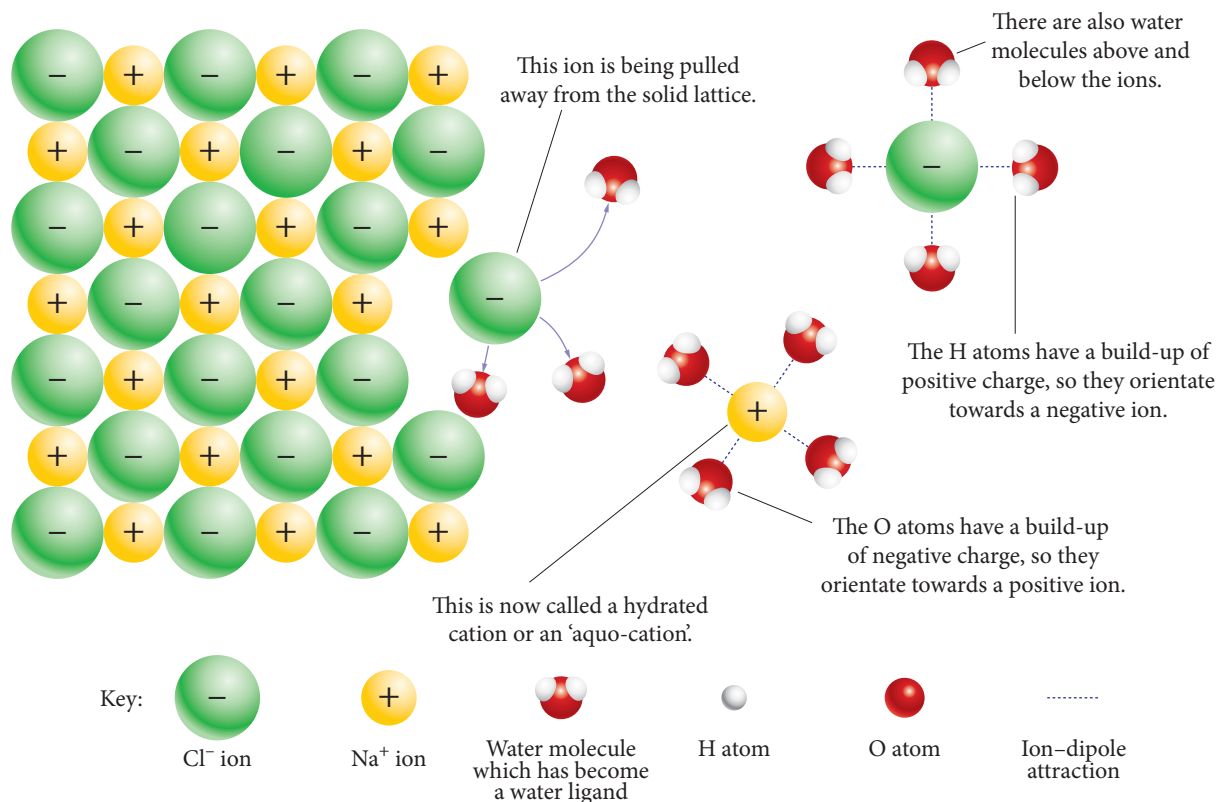
Water is considered to be the universal solvent. This is due to its polarity. Solutes are substances that dissolve in a solvent. The solution that is formed from a solute dissolving in a solvent is homogeneous (uniform), as the particles are too small to be seen.

Substances that readily dissolve in water are called **hydrophilic** ('water loving'). These include ionic compounds and polar substances such as some polar organic substances. Substances that don't dissolve in water are **hydrophobic** ('water hating'). These tend to be gases or non-polar substances.

Substances may dissolve in water if they are polar or charged. The solute may form an ion-dipole, dipole-dipole or hydrogen bond with water.

When a salt dissolves in water, ion-dipole bonds are formed. The positive part of the water molecule is attracted to the anion of the salt. More water molecules surround the anion until it is perfectly hydrated. The negative part of water is attracted to the cation and a similar process occurs until the ionic bonds within the salt crystal are broken and the salt has dissolved.

Water is polar and can dissolve polar substances by forming ion-dipole, dipole-dipole or hydrogen bonds with the substance.



▲ Figure C6.22

A simplified model of the solution process for a sodium chloride crystal in water



To revise how oil and water don't mix, refer to Context 3, 'Water, the vital substance', page 65.

Organic compounds as solutes

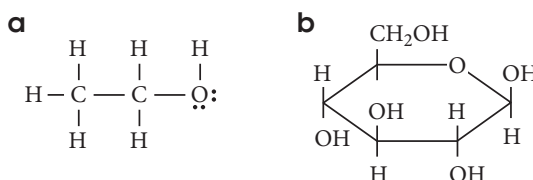
Most organic molecules are non-polar and hydrophobic. When polar groups are attached to the carbon chain, the hydrophobic nature of organic molecules decreases. Groups that give polarity are -OH , C=O and -NH .

If a large proportion of these polar groups is present, then organic substances such as glucose, amino acids, vitamins, enzymes and hormones may dissolve in water. As you discovered earlier, ethanol dissolves in water because it has a polar end due to the OH group. Because it also has a non-polar end due to the CH_3 , ethanol will also dissolve in organic solvents. Larger alcohol molecules that have longer carbon chains are not able to dissolve in water because the hydrophobic carbon chain predominates.

Non-polar substances dissolve in non-polar solvents but do not dissolve in polar solvents. Polar substances dissolve in polar solvents but do not dissolve in non-polar solvents.

Figure C6.23 ►

a) Ethanol and b) glucose molecules are polar due to the presence of polar regions.



ACTIVITY 6.3

DISAPPEARING ACT

Aim

To discover what happens when two liquids mix

You will need

- 2 accurately measured identical volumes of ethanol and water
- Large measuring cylinder

What to do

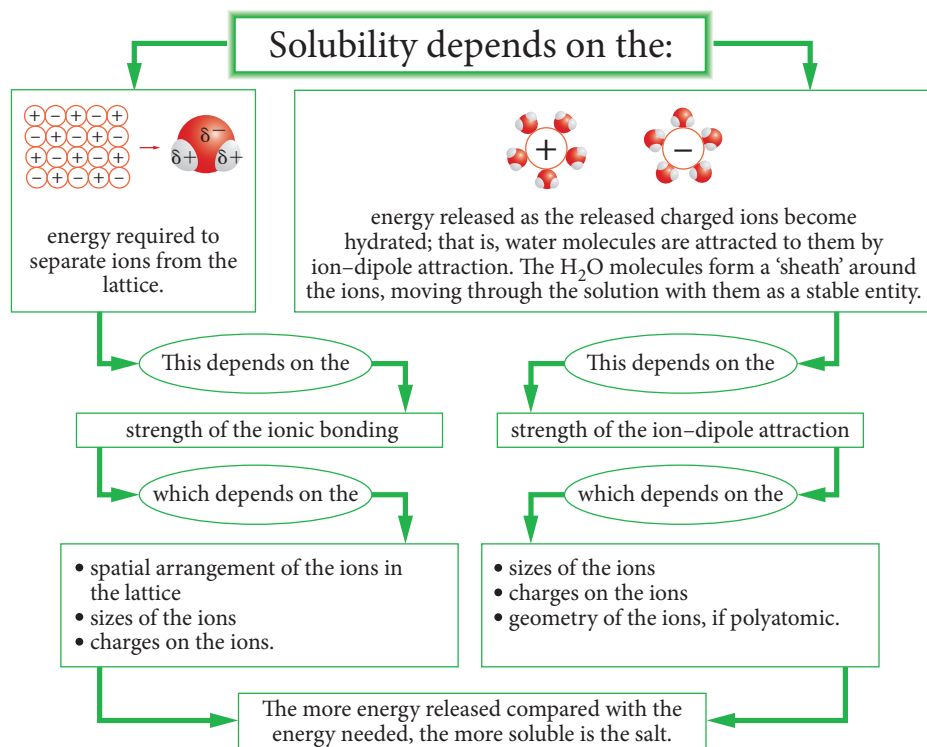
- 1 There are three options of what could happen on mixing the two liquids. The total volume could stay the same as the combined individual volumes, it could be greater or it could be less. For example, if 50 mL of ethanol and 50 mL of water are combined, is the final volume 100 mL, more or less? Predict what you think will happen.
- 2 Carefully pour the water and ethanol into a measuring cylinder.

What did you observe?

- 1 How many of the class accurately predicted the result?
- 2 Would you expect the same result if ethanol was poured first, or poured faster or slower. What causes any bubbles you may have seen? Can you repeat the experiment without bubbles?
- 3 Was this a chemical reaction? What must occur for it to be a chemical reaction? Did you have any evidence for this?
- 4 What interaction could occur between the molecules?
- 5 Summarise what must have happened as the two liquids mixed.

Other factors influencing solubility

For polar substances to dissolve in water, the energy required to separate the particles has to be less than the energy released when the particles are hydrated. The temperature of the solution alters both the rate at which a substance will dissolve and the amount that can dissolve. If a substance reacts with water, then the solubility is higher.

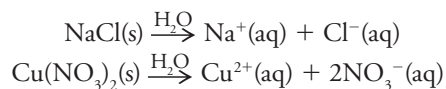


◀ **Figure C6.24**

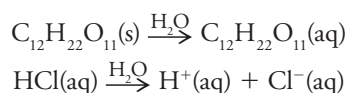
The role of energy factors in controlling the solubility of ionic compounds

Solutions equations (ionic and molecular)

You can tell a lot about what is happening in a reaction from the states in a chemical equation. When salts dissolve, chemists don't generally write the ions separately. For example, dissolved sodium chloride is written as NaCl(aq). The state aqueous (aq) indicates that the ions have **dissociated**. The Na⁺ and Cl⁻ ions are separated, and surrounded by water molecules, as you saw in Figure C6.22. As these are charged ionic particles, the solution can conduct electricity; it is an **electrolytic solution**. Two examples for the dissociation of salts are:



When molecular substances such as carbon dioxide, hydrochloric acid (HCl) or glucose (C₆H₁₂O₆) dissolve, the state is aqueous. When acids such as HCl ionise in water, they are written as HCl(aq). This means that the solution contains separate H⁺ and Cl⁻ ions.



The equations have to be balanced for charge and number of atoms of each element. The solid has no net charge, so the total combined charge of the dissociated ions has to be zero.

You will learn more about the reactions that occur when substances such as HCl or CO₂ dissolve in water in Chemistry Chapter 7.

After a substance becomes aqueous, further reactions may take place. Carbon dioxide has to dissolve in water before it can react with the water. The reaction with water is written separately. Oxygen has to dissolve for fish and other aquatic organisms to be able to use it.

The ability to dissolve in water may indicate that the substance has polar regions or a that reaction has occurred with water. Typical reactions are ionisation or dissociation reactions.

A solute dissolves when the energy of the bonds it forms with water is lower than the energy of the bonds between water molecules or between the water molecules and ions of the substance being dissolved. The energy difference explains why some polar salts dissolve in water and some don't.

QUESTION SET 6.4

Remembering

- 1 Draw a diagram that shows sodium chloride (NaCl) dissolving in water.
- 2 Write the formula for:
 - a lithium bromide (LiBr) crystallising as a dihydrate.
 - b calcium nitrate (Ca(NO₃)₂) crystallising as a tetrahydrate.
 - c magnesium nitrate (Mg(NO₃)) crystallising as a hexahydrate.

Understanding

- 3 Hydrogen bromide and methane are small covalent molecules. Explain why methane does not dissolve in water but hydrogen bromide does.
- 4 Describe what happens when sugar is added to a cup of coffee. Identify the solute and solvent.
- 5 KBr dissolves in water by dissociation.
 - a What is meant by this statement?
 - b Write an ionic equation to show what happens.
 - c Would this solution be called an electrolyte?
- 6 Describe what happens on the molecular level when water and ethanol mix. Explain by using the terms 'solute' and 'solvent'. What bonds are broken and what ones are formed? Write an equation that shows this.

Applying

- 7 Copy and complete the following table.

Bonding type	Solubility in water	Examples
Ionic	Most are soluble	a
b	Soluble if hydrogen bonds possible	Ethanol, glucose
	Soluble by reacting with water	HCl, HNO ₃ , NH ₃
	Otherwise insoluble	Dichloromethane
Non-polar molecular	c	O ₂ , I ₂
	Most are insoluble	d
Very large molecules	Insoluble if highly structured	e
	f	Starch, glycogen, enzymes
g	h	Diamond, SiO ₂
Metals	i	j
	Unless they react with water	Li, Na, K, Ca, Ba

8 Classify the following as either soluble in water or not soluble. Explain your reasoning.

- a HBr
- b MgCl_2
- c CS_2
- d ZnCl_2
- e H_2SO_4
- f SiCl_4
- g Glucose (a sugar, $\text{C}_6\text{H}_{12}\text{O}_6$)
- h C_8H_{18}
- i NH_2CONH_2

Analysing

9 A student was determining the water of crystallisation of magnesium sulfate. Unfortunately, he couldn't read his initial weights. In his rush to share the results with his prac partners he introduced errors.

His calculations and results are shown below. What are the errors and what was the initial weight for the anhydrous sulfate?

Mass of anhydrous magnesium sulfate = ? Dried weight = 1.65 g

Empirical formula:	$\frac{1.65}{120}$	:	$\frac{3.228}{138}$	convert mass to moles
	0.13749	:	0.02339	simplest ratio
	1	:	1.7	whole number ratio
	2	:	3	$2\text{MgSO}_4 \cdot 3\text{H}_2\text{O}$

Reflecting

10 Enzymes are denatured when their shape changes with high temperatures. However, some organisms can survive at temperature above 60°C . Enzymes in all organisms are made from the same building blocks (amino acids). What differences would be expected in the structure and bonding of these enzymes?

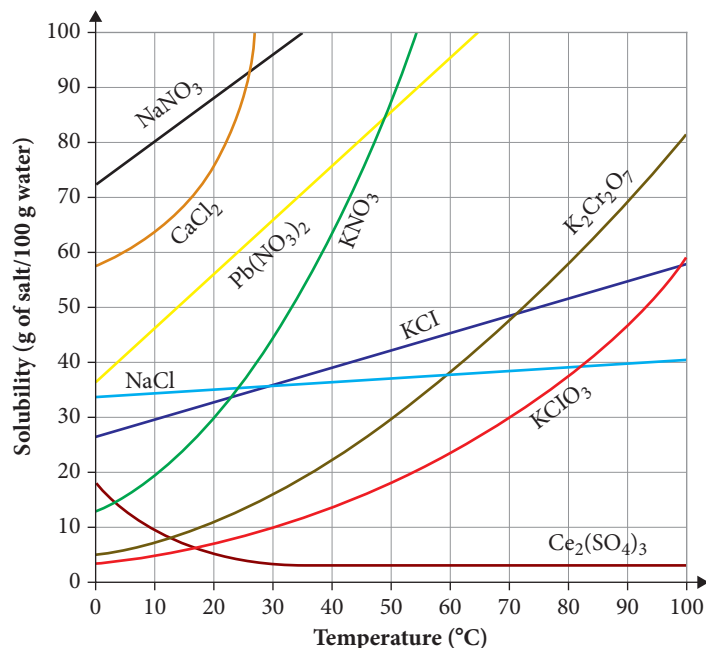
6.7 Measuring solubility

The amount of a substance that dissolves depends upon temperature. The **solubility** of a solute is the maximum amount in grams that can dissolve in 100 g of the solvent at a given temperature.

When adding sugar to a cup of water, there reaches a point at which no more sugar can dissolve at a particular temperature. The extra sugar stays at the bottom of the cup. The solution is said to be saturated with sugar. Before this point was reached, it was an **unsaturated solution**.

If the water is heated, then more sugar will dissolve. If this heated solution is allowed to cool, crystals will start to precipitate again. The temperature at which this happens is the temperature at which the solution is saturated. A solution can become supersaturated. This means that there is more solute dissolved than in a **saturated solution** at the same temperature. If this **supersaturated solution** is bumped, a sugar crystal is added or the side of the glass is scratched, then the extra sugar will precipitate out again. This is because the solution is unstable. Toffee is a supersaturated solution of sugar. To stop the toffee crystallising, a mixture of sugars or the addition of some fats can be used to stabilise the solution.





Solubility curves

A graph showing how much of a solute such as a salt can be dissolved at a particular temperature provides useful information on the solubility of a substance. Generally, as temperature increases more salt will dissolve. This graph is called a **solubility curve**, as shown in Figure C6.25. The amount of solute is measured in grams of solute in 100 grams of water.

◀ **Figure C6.25**

A selection of solubility curves for various salts

WORKED EXAMPLE 6.1

Refer to the solubility curve in Figure C6.25.

- How much potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) will dissolve at 20°C ?
- How much extra will dissolve at 80°C ?

Answers

- At 20°C , approximately 12 g of $\text{K}_2\text{Cr}_2\text{O}_7$ will dissolve in 100 g of water.
- The extra amount that would dissolve at 80°C is 46 g.

Logic

Find the curve for $\text{K}_2\text{Cr}_2\text{O}_7$ on the graph. Draw a line from 20°C up to the potassium dichromate curve. Then draw a line from this point across to the y-axis. This will indicate the amount that will dissolve at 20°C .

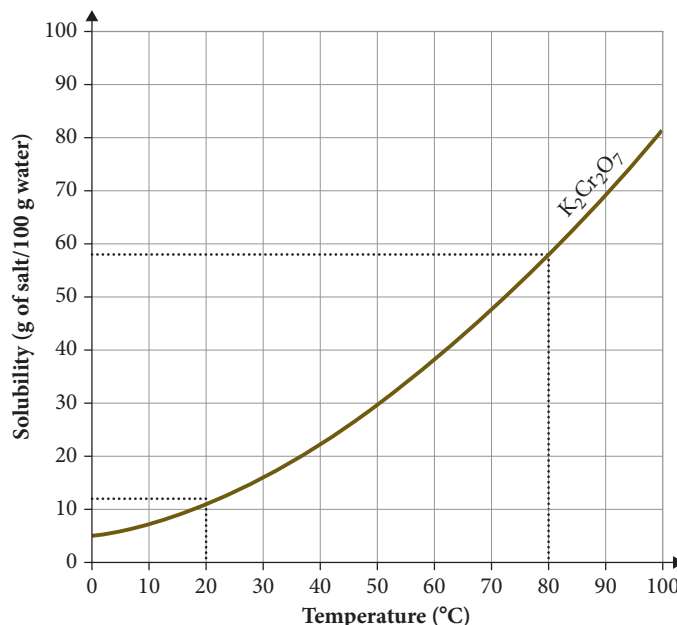
Similarly, if the temperature is increased to 80°C , approximately 58 g of the substance will dissolve in 100 g (or 100 mL) of water.

$$58 - 12 = 46 \text{ g}$$

This is also the amount that would crystallise out of solution if the temperature of the saturated solution was cooled from 80°C to 20°C .

Figure C6.26 ▶

Calculating the amounts of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) that can dissolve at 20°C and at 80°C . The difference is how much extra or how much potassium dichromate will crystallise out of the solution on lowering the temperature from 80°C to 20°C .



Try these yourself

- a What mass of the following solutes will dissolve in 100 mL of water?
- $\text{K}_2\text{Cr}_2\text{O}_7$ at 50°C
 - NaCl at 100°C
 - NaNO_3 at 10°C
- b Determine which of $\text{K}_2\text{Cr}_2\text{O}_7$, NaCl and NaNO_3 is most soluble in water at 15°C .
- c At 30°C , what mass of KCl and NaCl would just dissolve in 100 mL of water. For this mass at this temperature, would the solution be saturated, unsaturated and supersaturated?

The solubility of a substance in a particular solvent is the maximum amount of that substance that can dissolve at a particular temperature.

A saturated solution is one in which no more of a particular solute can dissolve in a given quantity of solvent. A solution with less than this amount is an unsaturated solution.

A supersaturated solution is unstable and has more than the maximum amount of solute dissolved in a given quantity of solvent. It has a higher amount of solute than the saturated solution.

EXPERIMENT 6.2

DERIVING THE SOLUBILITY CURVE OF AMMONIUM CHLORIDE

Aim

To gather data to draw a solubility curve of ammonium chloride

Materials

- 250 mL beaker
- 10 mL measuring cylinder
- Large test tubes
- Bunsen burner
- Tripod
- Retort stand
- Bosshead and clamp
- $0\text{--}100^\circ\text{C}$ thermometer
- Stirring rod
- Distilled water
- Balance
- Weighing bottle
- Safety glasses

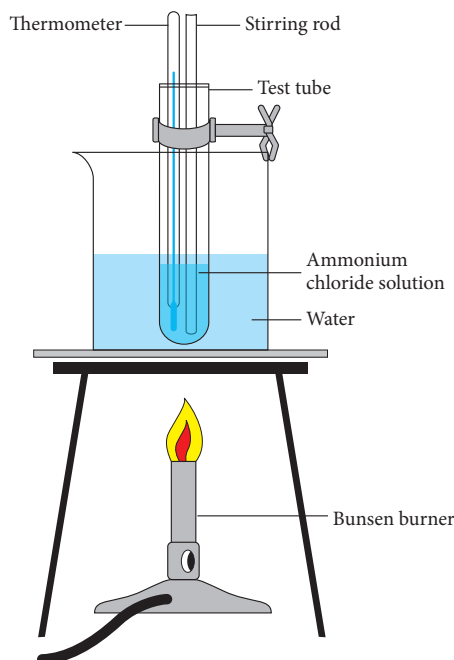
What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Ammonium chloride is slightly toxic.	Work in a well-ventilated area. Wear safety glasses and wash your hands after the experiment. Dispose of all chemicals according to your teacher's directions. Do not pour them down the sink.
The Bunsen burner will get hot.	Do not use the Bunsen burner if the gas tube is damaged. Ensure long hair is tied back and the flame is away from flammable material. If you burn yourself, place the affected part under cold running water for 10 minutes and inform your teacher.
Broken glassware will cut.	Inspect and discard any chipped or cracked glass wear, no matter how small the damage. Sweep up broken glass with a brush and dustpan; do not use your fingers.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

- 1 Carefully weigh out the amount of ammonium chloride assigned to your group.
- 2 Place the weighed ammonium chloride into a test tube.
- 3 Add 10 mL of distilled water.
- 4 Half-fill the beaker with tap water, clamp the test tube securely into the beaker so that it is immersed in the water.
- 5 Warm the beaker, stirring the contents of the test tube with the stirring rod. Continue until the ammonium chloride has dissolved completely.
- 6 Allow the test tube to cool. Continue to stir it until tiny flakes of ammonium chloride start to appear. Note the temperature at which this occurs.

► **Figure C6.27**
Experimental set-up for determination
of a solubility curve



Results

Copy and complete the following table.

Group	Mass of ammonium chloride (g)	Mass of water (g)	Temperature of recrystallisation (°C) (x-axis)	Solubility (g/100 g H ₂ O) (y-axis)
1	4.00	10.00		
2	4.50	10.00		
3	5.00	10.00		
4	5.50	10.00		
5	6.00	10.00		
6	6.50	10.00		

Analysis of results

- 1 Use the class results to plot a graph of solubility against temperature. Plot the temperature along the horizontal x-axis (0–100°C).
- 2 From the graph, predict the solubility of ammonium chloride at 20°C, 40°C, 60°C and 80°C.

Discussion

- 1 Describe what happens to the solubility of ammonium chloride as temperature increases.
- 2 At the point the crystals first appear, the solution is saturated. What does 'saturated' mean?
- 3 At what temperature would 10 g of ammonium chloride saturate 50 g of water?
- 4 If a saturated solution of ammonium chloride at 90°C is cooled to 10°C, how many grams of ammonium chloride would crystallise out of 100 g of water?
- 5 If the theoretical value for the solubility of ammonium chloride at 50°C is 50 g/100 g, what percentage error does your experiment have?

$$\text{Percentage error} = \frac{\text{experimental value} - \text{true value}}{\text{true value}} \times 100$$

- 6 List possible sources of errors in your experiment.

Taking it further

Describe modifications to the experiment that could improve the accuracy.

Gas solubility curves

The solubility of a gas can also be shown with a solubility curve. The solubility curve for oxygen and carbon dioxide was shown in Context 3 (page 71). As temperature increases, the solubility of non-polar gases generally decreases. Solubilities are generally much lower than for solids and typically the values are shown in parts per million (ppm).

Concentration units

Solubility curves use the unit grams of solute per 100 g of water. This can also be called % w/w. If you are weighing the amount of salt added to 100 g of a food, then the % w/w is the easiest to use.

Two other similar units use volumes (% v/v) or a combination of weight and volume (% w/v).

The units you use depend on what you are measuring. For example, you may wish to investigate the alcohol content of drinks. Ethanol is a liquid, so it is easier to measure its volume than its weight. The unit to use is % v/v. The alcoholic content of wine, beer and spirits is measured in volume of ethanol in 100 mL of drink. Note that the volume is 100 mL not 1 litre, which is the standard volume of measurement in chemistry. Having a different unit means that we can use percentage values. This is useful for different-sized containers such as a 750 mL or 1.25 L bottle.

In section 1.9 on page 127, you looked at ppm for a calibration curve of atomic absorption spectroscopy. The unit ppm is very useful when referring to the amount of the solute at very low levels.

In section 7.2 on page 312, you will learn about another way to measure concentration that uses moles and molarity.

Concentration is a measure of the amount of solute present. There are a variety of units in general use; g/100 g, % v/v (per cent by volume), % w/w (per cent by weight), % v/w, ppm (parts per million, grams of solute in one million grams of solution.)

WORKED EXAMPLE 6.2

- a** Red wine has an alcohol content of approximately 13 v/v %. In a 750 mL bottle, what amount of alcohol is present?
- b** A hair sample was found to have a copper level of 41 ppm. It should be less than 2.5 mg % w/w. Is this an acceptable level?

Answers

- a** 97.5 mL of ethanol in 750 mL of red wine

Logic

$$\begin{aligned} &\text{Total volume} \times \text{percentage alcohol} \\ &750 \text{ mL} \times \frac{13 \text{ mL}}{100 \text{ mL}} = 97.5 \text{ mL} \end{aligned}$$

- b** No; 41 mg % w/w is too high.

- 1** Convert to an amount in grams.

$$\begin{aligned} 41 \text{ ppm} &= 41 \text{ g per } 10^6 \text{ g} \\ &= \frac{41 \times 100}{10^6} \\ &= \frac{41}{10^4} \end{aligned}$$

- 2** The legal limit has the % w/w as a mg value so convert to milligrams.

$$\begin{aligned} &= 41 \times 10^{-6} \times 100 \text{ \% w/w} \\ &41 \times 10^{-6} \times 100 \times 1000 = 4.1 \text{ mg \% w/w} \\ &\text{No, level is too high.} \end{aligned}$$



Try these yourself

- a 200 mL of a sport drink has 6% w/v of sugar. What amount of sugar is present?
- b A 300 mL glass of champagne contains 36 mL of alcohol. What is the concentration (v/v) of alcohol?
- c A solution has a cadmium ion level of 500 ppm. How much cadmium would be present in 100 mL of this solution?

It is helpful to remember that
 $\text{ppm} = \text{mg L}^{-1} = \mu\text{g g}^{-1}$

ACTIVITY 6.4

SPORT DRINKS: HOW MUCH SALT CAN YOU DRINK?

Too much salt in our diet can cause health problems. Salt is a common additive in many foods and drinks. Sport drinks are designed to replace salts lost in sweat. If the salt level in the drink is too low, then the drink is hypotonic. If the salt level is too high, then the drink is hypertonic.

Aim

To determine the minimum amount of salt that you can detect in a glass of water.

You will need

- 6 clean glasses
- Table salt (NaCl)
- Electronic balance
- Tap water
- Spoon
- Bucket or waste container
- Marker pens or coloured stickers

All equipment must be food grade and not the general laboratory supplies.

What to do

- 1 Do not perform this activity in a laboratory because it requires you to taste the samples.
- 2 Prepare six separate 500 mL water samples that have a range of salt levels from 0 to 5 g of added salt. For a comparison, brackish water has 3 g L^{-1} and sea water 35 g L^{-1} . Ensure that the salt is completely dissolved in each sample. Use marker pens or stickers to distinguish the samples.
- 3 Pour small amounts of each sample into glasses. Give the samples to your partner in order from no salt to most salt or from high salt to low salt. Make sure that the amounts that they consume are very small, typically less than a few millilitres. Have fresh water to drink in between if required. Do not swallow the sample but spit it into the bucket or waste container for disposal.

What did you discover?

- 1 Did different people have different detection limits?
- 2 Do you think that people were influenced by what they observed or the order they were given?
- 3 Which samples could you drink readily?
- 4 Convert the amounts in your samples to % w/v and ppm.
- 5 Why is the unit % w/v better to use than % w/w in this case?

Taking it further

- 1 Repeat the experiment with sugar. What is the minimum of sugar that can be detected?
- 2 Research why and how manufacturers chase the 'bliss point' in food.

QUESTION SET 6.5

Remembering

- 1 When a solvent holds as much of a solute as it normally can at a given temperature, we say the solution is _____. When a solvent has less dissolved solute than it normally can at a given temperature, we say the solution is _____.

Applying

- 2 Samples of brackish waters were found to have 30 g L^{-1} of salt. How much salt can be obtained from:
 - a 50 mL?
 - b 300 mL?
- 3 Bottles containing 40 mL of brand X, a liquid plant food, contains 3.2 g of seaweed extract. What is the percentage concentrate of seaweed extract in the plant food?
- 4 The fluoride concentration of drinking water should be less than 1 ppm on a mass to volume basis. A 500 mL sample of water was found to contain 0.3 mg of fluoride ions.
 - a Calculate the concentration of fluoride ions in the water in % w/v and parts per million.
 - b Is this below the recommended level?
- 5 How many grams of potassium nitrate would be needed to saturate 50 g of water at 20°C ?
- 6 Sodium thiosulfate will dissolve in water readily. At 25°C , a saturated solution will have 120 g of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ for each 100 g of water.
 - a You take a test tube and add a crystal of sodium thiosulfate to 2 cm^3 of water. Is it likely to be saturated?
 - b You take another test tube and add a similar amount of water and then add crystals until no more dissolves. What term describes this?
 - c In a third test tube, you add 2 cm^3 of crystals and a few drops of water. Why would you not expect all the crystals to dissolve?
 - d You heat it until the solution is clear and then cool it back to room temperature. What would happen if a crystal were now added to the test tube?
- 7 A 20 g solution of a pesticide contains 0.5 g of active ingredient. What is the concentration in % w/w and ppm.
- 8 On the bottle of every sport drink is a label detailing the amount of salts and sugars present, usually in 1 litre of product. Compare sport drinks compositions. If you had to make 750 mL of a sport drink, what and how much would you add to 750 mL of water?
- 9 A 200 mL sample of tap water was found to contain 0.2 mg of fluoride ions.
 - a Calculate the concentration of fluoride ions in the water as ppm.
 - b As the limit should be between 0.7 and 1.2 ppm, does this concentration exceed the recommended limit of 1 ppm?
 - c The recommended maximum daily intake of fluoride is about 10 mg/day. How many cups would a person need to drink to reach this limit? Assume that water is the only source of fluoride.

Reflecting

- 10 A gluten-free diet is considered to be the only treatment for people with coeliac disease. Australian manufacturers of gluten-free products must ensure that no detectable amounts of gluten are present. Manufacturers are pushing the regulators to make Australian standards the same as those in Britain and Europe – a maximum of 20 mg of gluten per kilogram. Why is this so important for a person with coeliac disease?



For some examples of the benefits of the practical applications of water, refer to Context 3, 'Water, the vital substance', pages 66 and 69.

6.8 Practical applications

Many practical uses of water rely on its high surface tension and ability to act as a solvent. These applications rely on the surface interaction and intermolecular forces between polar and non-polar substances.

Surfactants: polar and non-polar

Oils and water don't mix; non-polar liquids will not mix with polar solvents. These two substances are **immiscible**; they are unable to be mixed. When a **surfactant** is added to the mixture, the tension between the oil and water is reduced. They appear to dissolve in each other. The result is an **emulsion**, a mixture of the two immiscible substances. The compound that helps form the emulsion is called an emulsifier. Eggs are a good emulsifier for oil and vinegar. The resultant emulsion is called mayonnaise. Milk is considered to be an 'oil in water' emulsion. When whipped sufficiently, it turns into a 'water in oil' emulsion, called butter. The excess water is separated in the process.

ACTIVITY 6.5

EMULSIONS, COLLOIDS AND SUSPENSIONS

When substances are added to a liquid they do not always form a solution. They may form an emulsion, a **colloid** or a **suspension**.

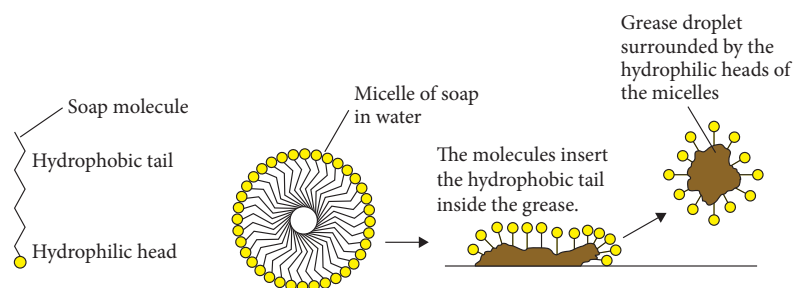
- 1 What distinguishes these three mixtures?
- 2 What are the common uses of these mixtures? Give examples that you may see around your home.

Detergents: surfactants that clean

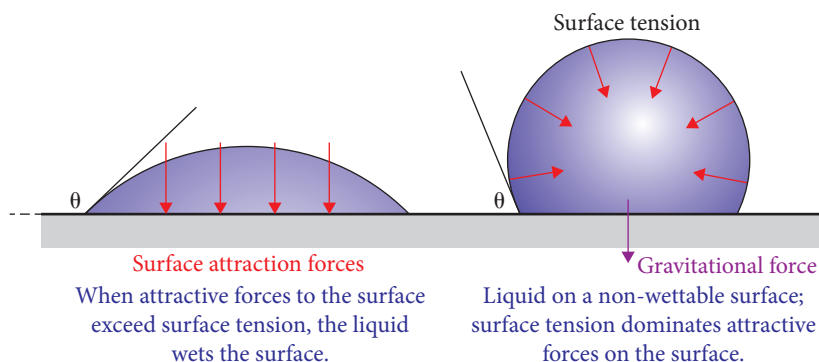
Soaps and detergents are everyday examples of surfactants. One end of a surfactant is hydrophilic and the other is end hydrophobic. Soap has a polar head (typically a sodium ion) attached to a non-polar long hydrocarbon chain tail. Soap forms **micelles**, sub-microscopic clusters in water, which have polar ends on the outside and hydrophobic tails in the middle where they can avoid water. The long tail of the soap dissolves into the grease because of dispersion forces. There is an ion-dipole interaction between the polar head and the water molecules. More and more hydrophobic tails of the soap molecules will tunnel into the grease droplet, eventually incorporating it into the water.

Figure C6.28 ►

Soap molecules interacting with grease. Over time, more and more soap molecules are incorporated into the grease molecule. Eventually, a small grease droplet is surrounded by the hydrophilic heads of the soap molecules, forming a micelle.



The surface tension of water can be lowered by dissolving surfactants such as soap or detergent. When present, the surfactant breaks the hydrogen bonding between the molecules and the water spreads out onto the surface (wets the surface) rather than remaining in a droplet.



◀ **Figure C6.29**

Wettability: a liquid will wet a surface if there is strong intermolecular attraction between the liquid and the surface. A liquid will bead on the surface if there are stronger cohesive forces within the liquid than the adhesive forces between liquid and the surface. The contact angle (θ) is a way of measuring the wettability.

ACTIVITY 6.6

WETTABILITY

Aim

To determine the difference in bonding of water to various substrates such as plastic, glass and nanotechnology cloth and how this bonding is affected by detergents

You will need

- Dropper
- Clean glass
- Plastic
- Water-resistant nanotechnology material
- Detergent

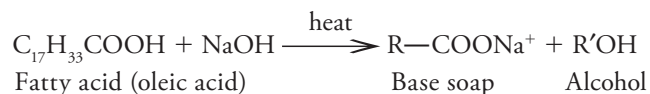
What to do

- 1 Place a few drops of water onto each of the surfaces and notice the shape of the droplet.
- 2 Add a single drop of detergent to each drop of water.
- 3 Record what happens.

What did you discover?

Describe what happened when you added the detergent to each drop.

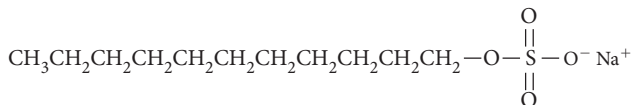
Soap is one of the oldest surfactants. It is made from fatty acids in animal and vegetable oils. The sodium or potassium salts of these fatty acids have the general formula RCOO^-Na^+ , where R represents a long carbon chain. Sodium stearate is a soap made from the fat of cattle or sheep reacting with a strong base such as NaOH. The **saponification** reaction is:



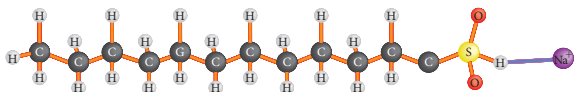
The soap formed has a polar head (COONa^+) and a long non-polar hydrocarbon tail (R). The polar head is strongly attracted to the water molecules and the long tail is hydrophobic. The tail

will dissolve into the grease or oil. Detergents are used in **hard water** rather than soap, which does not form a lather. Detergents have an ionic head based upon sulfate ions. These sulfate ions do not precipitate the calcium or magnesium ions in hard water as soap does. Detergents are classified according to the charge on their polar head (Figure C6.30). **Anionic** detergents have a negatively charged sulfate ion. **Cationic** detergents have a positively charged end. **Non-ionic detergents** do not have any ionic groups.

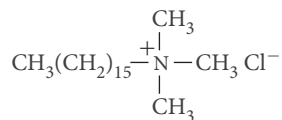
Anionic detergent



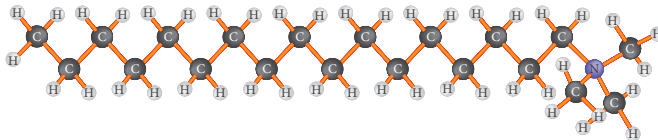
Sodium lauryl sulfate



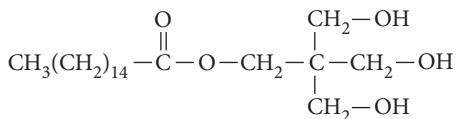
Cationic detergent



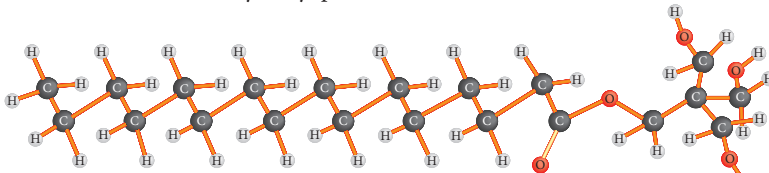
Trimethylhexadecylammonium chloride



Non-ionic detergent



Pentaerythrityl palmitate



Non-polar

Polar

▲ Figure C6.30

The three types of detergent: anionic, cationic and non-ionic. What enables these molecules to dissolve in water?

No need to bathe or wash again?

Nanotechnology offers many advantages in a world becoming more concerned about water availability. By changing the surface of materials, there may be no need to clean them. Sweat or dirt will just not stick. A surface that beads the water, which then runs off, offers advantages to the builders of high-rise buildings, who may be daunted at the prospect of cleaning windows 40 floors up. Car paint companies are looking at applying nanotechnology to the outside of cars to remove the need to clean the car. Clothing companies are some of the biggest users of nanotechnology.

Surfactants such as detergents interrupt the surface tension of water. Surfactants have both polar and non-polar regions.

6.9 Chromatography

Knowledge of how solutes dissolve in solvents and interact with the surface of materials is the basis of the analytical techniques of chromatography. There are three main techniques: thin-layer chromatography, gas chromatography and high-performance liquid chromatography.

The chemical basis for these techniques is that different substances will **adsorb** onto a surface and **desorb** into a solvent at different rates. This difference is due to many factors of the substance being tested (**analyte**), including:

- the different types of polar groups
- the amount of charged and polar chemical groups present
- its molecular weight
- its geometry
- the positions and numbers of carbon–carbon double bonds.

By altering the solvent (**mobile phase**) and the surface (**stationary phase**), chemists can separate a mixture of solutes.

Chromatography is the name of the group of techniques that separate substances based upon differential distribution between a stationary phase and a mobile phase.



To see applications of chromatography in forensic testing, refer to Context 1, 'Matter in the universe', page 11.



To see applications of chromatography in blood and water testing, refer to Context 3, 'Water: the vital substance', pages 73 and 74.

INVESTIGATION 6.2

CHROMATOGRAPHY OF PLANT PIGMENTS

The plant pigment chlorophyll absorbs the light energy used in photosynthesis. When trying to discover how chlorophyll works, chemists had to first isolate chlorophyll, find out how many different types there are and determine whether all plants had the same pigment.

Aim

To separate plant pigments using chromatography

What will you need?

You will need to design a method to separate the plant pigments in plants by chromatography. Hint: Plant pigments will dissolve best in an ethanol and water mixture. Filter paper or chalk can be used to separate the pigments. Sand can help to break open the cells by grinding the plants. Calcium carbonate can neutralise acid released from the cells as they are crushed.

Complete the risk assessment to show any risks in undertaking this investigation and ways to reduce them. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?
Heating a mixture of ethanol and water will cause the beaker to get hot.	If you burn yourself, place the affected part under cold running water for 10 minutes and inform your teacher.
Ethanol is a toxic and flammable liquid.	Only directly heat solutions that have less than 50 % v/v of ethanol. Keep bottles away from flames and heat sources. Do not come into contact with the liquid or breathe the fumes.
Plants may be poisonous.	
Glassware	
Filter paper	

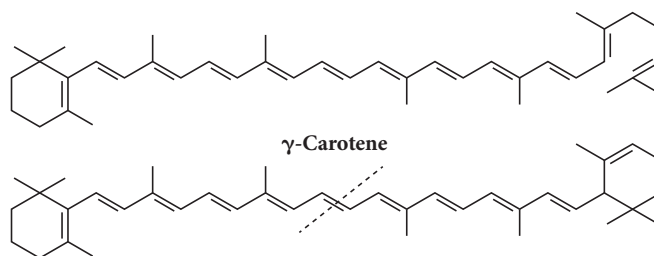
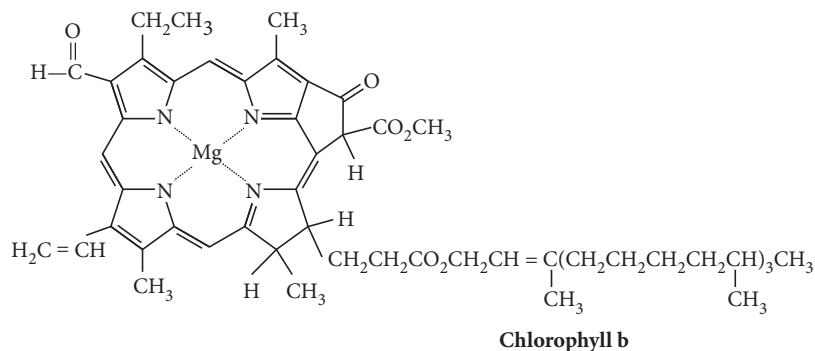
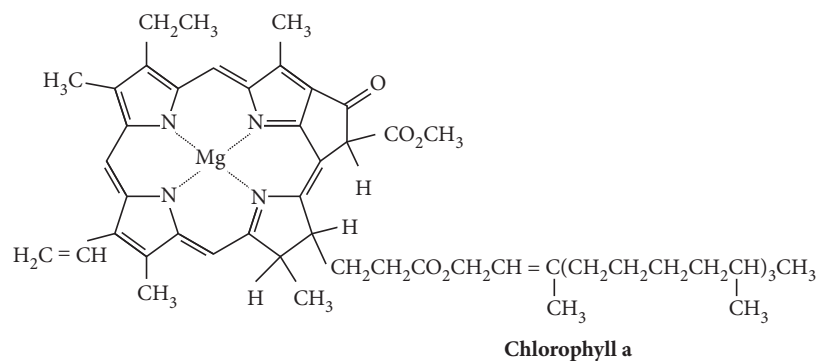


Figure C6.31 ►

Pigments in plant cells absorb light energy. There are many different pigment molecules, each a different colour. Four are shown here.

Questions to get you started

- 1 How will you carry out your investigation? Which method will you try first?
- 2 Will large pieces of plant interfere with your technique? If so, what should you do?
- 3 How long will you boil the plants for? How will you judge whether you have achieved the maximum extraction?
- 4 How will you use the filter paper or chalk to separate the pigments from the solvent?
- 5 How fine will the plant pieces have to be? What factors would increase the amount of pigment released?
- 6 How will you analyse your results?

After the experiments

- 1 How well did you manage to extract plant pigments?
- 2 Did other students succeed in extracting plant pigments? Compare the effectiveness of their method with yours.

Ideas for improvement

- 1 What were some of the problems encountered? What errors were encountered?
- 2 How would you refine this method? How will the refined method provide more consistent results?
- 3 If you suspected that one of the components was chlorophyll, how would you confirm this?

Thin-layer chromatography

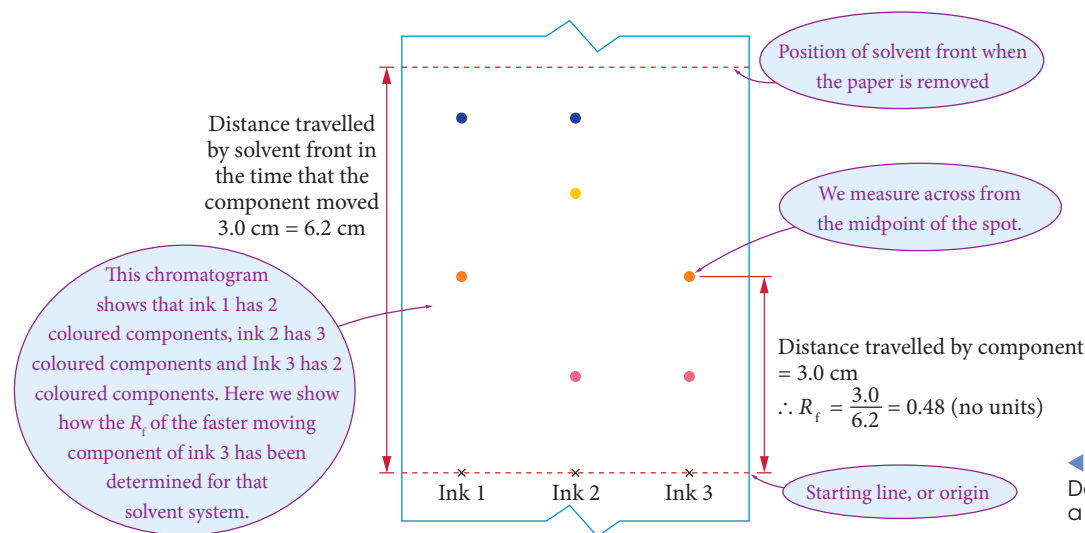
Thin-layer chromatography (TLC) is used to separate and analyse a wide variety of molecular mixtures. It is used to analyse the presence of particular drugs and amino acids; TLC is faster and provides more separation than paper chromatography.

The TLC plate typically consists of a 0.1-mm-thick layer of absorbent material bonded to a glass or plastic support. The absorbent material consists of many microscopic plates. These surfaces provide a large surface area for chromatographic separation. This is the stationary phase. The **origin** line is drawn in pencil about 1 cm from the base. This is where the samples are placed.

Separation is achieved by the solvent (mobile phase) moving up the stationary phase. This movement is due to capillary action. The solvent moves through the samples towards the top of the plate. The faint wavy line seen is called the solvent front. The samples will desorb into the solvent and travel up the plate with the solvent. If all components spent the same amount of time in the solvent, then they would all travel at the same rate. They don't because components have differences in chemical structure, which alters the adsorption and desorption rates. The particles most attracted to the stationary phase move the slowest. The particles that are most strongly attracted to the solvent move the fastest.

In TLC, you measure the distance the spot has moved from the origin compared to the distance the solvent has moved from the origin. This ratio is called the retardation factor or R_f . The R_f value must be between 0 and 1 and is defined as the distance travelled by the sample from its origin divided by the distance of the solvent front from the sample origin. A substance that does not migrate from the sample origin has an $R_f = 0$, while one that is not adsorbed at all (migrated with the front) has a $R_f = 1$. The R_f value is characteristic for a particular substance with a given absorbent and solvent system. The R_f value cannot be greater than 1.

The separation of the components depends upon the length of time the plate is in contact with the solvent. If this is too short, then there is only a small separation. If this is too long, then all the spots would eventually make it to the top. A longer plate and a reasonable time will provide the best separation. It is not always possible to identify all the components of a sample with one TLC plate. Some samples may have very similar R_f values. Different solvent mixtures and stationary phases can be tried to achieve better separation.



◀ **Figure C6.32**
Determining the R_f from a chromatogram

The proportion of time an analyte spends adsorbed compared with desorbed is consistent for that mobile and stationary phase. A material with an R_f of 0.5 will move 2.5 cm when the solvent front has moved 5 cm. If a second plate was allowed to run for longer, then the same sample will move 5 cm when the solvent has moved 10 cm. Therefore, you can consider the samples to be the same when the R_f values are the same.

You can confirm the identity of an unknown compound by running a pure sample of the compound (a standard) under the same conditions. If the substances are the same, then the R_f values should be the same.

EXPERIMENT 6.3

THIN-LAYER CHROMATOGRAPHY

Thin-layer chromatography (TLC) is an invaluable technique used in chemistry and biochemistry for the separation and analysis of a wide variety of molecular mixtures.

In this experiment, two different solvent systems will be used to separate a mixture of dyes on a TLC plate. The solvent should be allowed to move up about half the distance of the plate. The dye molecules contain different types and amounts of charged and polar chemical groups. They also have different molecular weights, geometry and positions and numbers of carbon-carbon double bonds.

After the experiment is completed and the plate is partially dry, a faint wavy line can be observed at the last location of the leading edge of the solvent in the absorbent stationary phase.

Aim

To use TLC to separate mixtures of dyes to determine their composition

Materials

- 2 TLC plates (5 cm × 10 cm)
- 5 different-coloured food dyes
- Distilled water
- Methylated spirits
- 2 × 250 mL beakers
- 5 capillary tubes
- Pencil

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Methylated spirits is a toxic, flammable liquid.	Keep bottles away from flame and heat sources. Do not come into contact with the liquid or breathe the fumes.
Glassware	

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each. Ask your teacher to check your risk assessment before you proceed.

Procedure

- 1 Measure 1 cm up from an edge on each plate and lightly draw a straight line across with a pencil. Do not press hard or you will scrape the cellulose off its support. This line will be the sample origin.
- 2 Place the end of a capillary tube into one of the dyes. Allow the liquid to move into the tube and then remove it from the dye.
- 3 Hold the tube vertically and touch the end of the tube to the pencil line, allowing a small amount of dye to be adsorbed onto the film.
- 4 Repeat this for the second piece of film.
- 5 Repeat steps 2 and 3 for different dyes. Use a new capillary tube for each different dye.
- 6 Pour distilled water into one of the beakers to a depth of approximately 5 mm.
- 7 Place the bottom edge of one of the plates in the beaker. Make sure the water level is below the pencil line.
- 8 Repeat steps 6 and 7 with methylated spirits.
- 9 Allow the plates to stand in the solvent for approximately 10 minutes.
- 10 Withdraw the plate, place it on a piece of paper towel and, using a pencil, mark the solvent front.
- 11 Measure the distance from the pencil line to the solvent front for each solvent and record this measurement.
- 12 Measure the distances each of the dye components has travelled from the origin in each solvent and record this in a results table.

Results

Copy and complete the following results table.

Dye	Distance for distilled water	R_f	Distance for methylated spirits	R_f

Analysis of results

Calculate the R_f values for each of the dyes and their components by dividing the distance travelled by the dye component by the distance travelled by the solvent.

Discussion

- 1 Compare the R_f values of the dyes in both solvents.
- 2 Which dye was adsorbed most strongly in:
 - a water?
 - b methylated spirits?
- 3 Which solvent was the best for separating the dyes?
- 4 TLC can be one of several methods used to help identify an unknown compound. Explain how this might be accomplished using the dyes as an example.

Gas chromatography

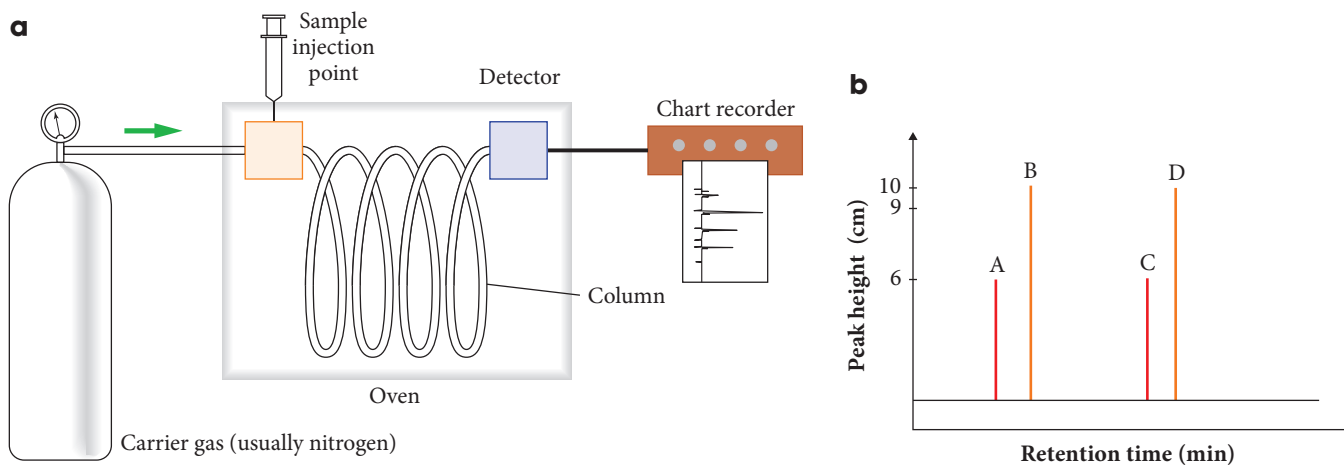
Gas chromatography (GC) is a separation technique for small organic molecules that can withstand relatively high temperatures. Blood alcohol levels are measured by gas chromatography.

The gas chromatograph consists of a gas bottle, an oven, a column, a detector and a recorder. The sample is injected into the oven where the gas pushes the sample into the long, thin column. In the column, smaller particles and those that adsorb onto the stationary phase the least leave the column first. Larger particles and those that adsorb more readily take longer to leave the column.

The time the sample takes to **elute** (leave the column) is characteristic for the substance. This time is called the R_t , the retention time. The R_t value is used to identify the component, as R_f is in TLC and paper chromatography.



To revise calibration curves, refer to Chemistry section 1.9 on page 127.



▲ Figure C6.33

A schematic diagram of: a) a gas chromatograph and b) a hypothetical gas chromatogram



LEARN MORE ABOUT HPLC

Watch this video to learn more about HPLC.

This technique can also measure the amount of the analyte. If a wine is thought to contain 12% alcohol, then a series of standards with alcohol ranging from 5% to 20% is run through the gas chromatograph and the peak areas are measured. A calibration curve is drawn and the unknown value is read off. It is important that the unknown value lies within the range of the standard values.

The R_f value will be the same for each standard. What is different is the height of the peak. The peak area will alter in proportion to the amount of the substance present. A calibration curve is plotted and the unknown wine's peak area is used to determine the amount of alcohol present.

High-performance liquid chromatography

High-performance liquid chromatography (HPLC) is used for larger organic molecules. It can be used for substances that are unstable to heat because there is no oven. The stationary column is shorter than in the gas chromatograph. The mobile phase is a liquid not gas. A pump is used because pressure is required to move the liquid through the column. As with GC, R_f values are used to identify compounds, standards are run to confirm the identity (same R_f) and a calibration curve is used to determine the amount of the substance present.

Figure C6.34 ▶

A schematic diagram of: a) a high-performance liquid chromatograph and b) a high-performance liquid chromatogram of pigments extracted from a *Dianella* plant

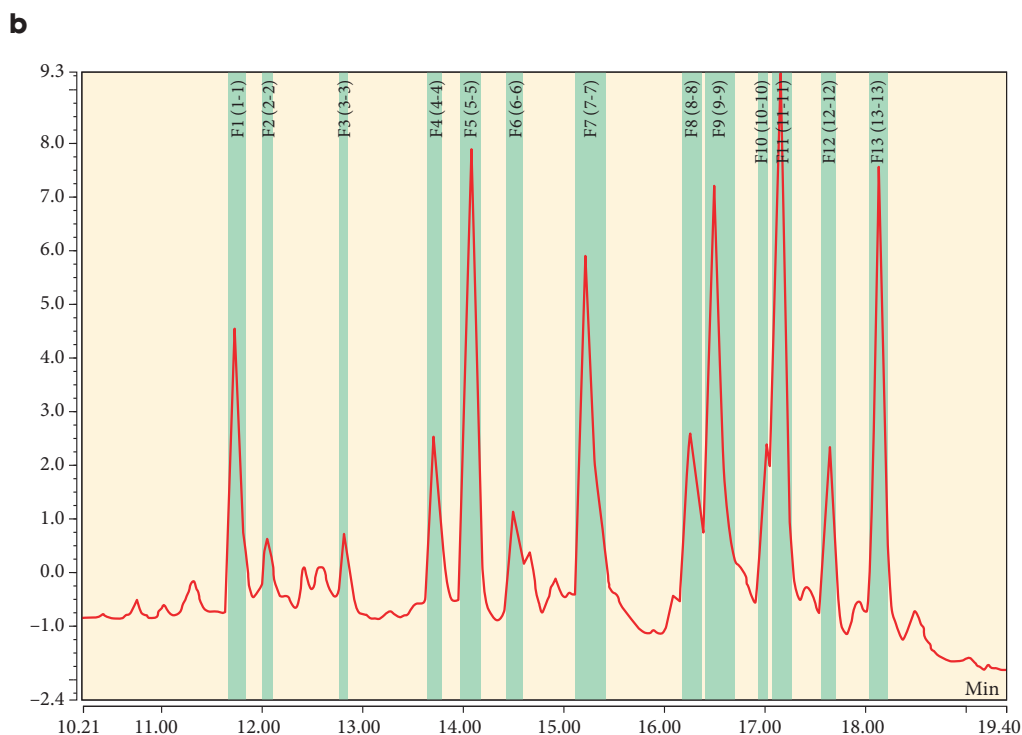
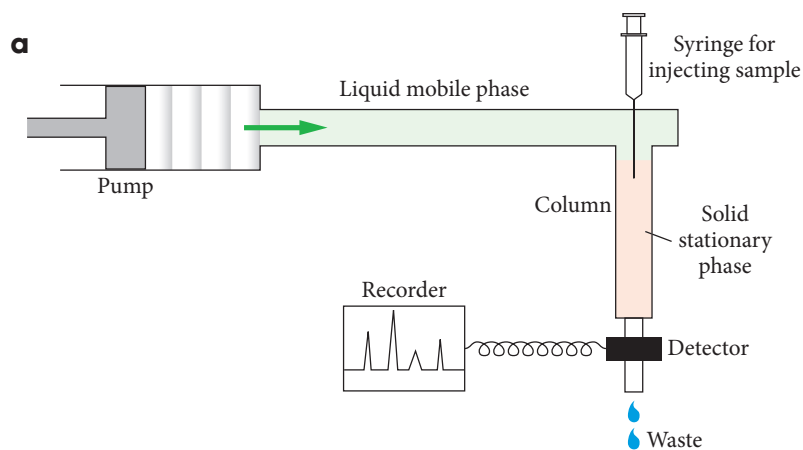


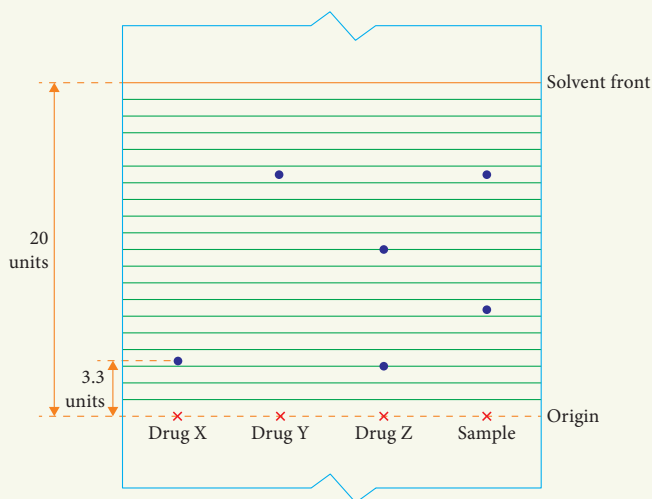
Table C6.3 Summary of chromatography techniques

Technique	Basis	Stationary phase	Mobile phase	Measures	Uses
Gas chromatography	Carrier gas is mobile phase. Sample is injected into oven. The gas moves the vaporised particles through a fine long column. The smallest solutes that adsorb the least, elute first. R_f depends on temperature, length of column and flow rate as well as chemical structure	Long, thin column	Inert gas such as N_2 , CO_2 or He	R_f indicates what the sample is. Area under peak indicates amount of substance present. Standards are used to confirm identity (qualitative) and to prepare a calibration curve for calculations of amounts (quantitative analysis)	Small, heat-stable organic compounds such as ethanol
High-performance liquid chromatography	Similar to GC but mobile phase is a liquid. Unlike the gas in GC, the liquid plays an important role in adsorption and desorption. Needs a pump to push the liquid through the denser packed column	Short column (particle size of stationary phase is extremely small, allowing better separation)	Water-methanol or water-acetonitrile mixes	R_f indicates what the sample is. Area under peak indicates amount of the substance present. Standards are used to confirm identity (qualitative) and to prepare a calibration curve for calculations of amounts (quantitative analysis)	Larger organic compounds
Thin-layer chromatography	Solvent moves over stationary phase by capillary action. Components adsorb and desorb at different rates due to chemical structure and ability to form intermolecular bonds to the stationary phase and mobile phase.	Fine powder on glass or plastic	Solvent may be organic or water mixture	R_f is characteristic for each sample at the same conditions. Standards are used to confirm identity (qualitative)	Organic mixtures such as plant pigments and drugs

QUESTION SET 6.6

Remembering

- 1 List the differences between soap and a detergent. Describe why detergents work with hard water but soaps don't.
- 2 Figure C6.35 represents a thin-layer chromatogram of drug samples confiscated by the police. Included on the plate are commercial drugs, labelled X, Y and Z.



◀ Figure C6.35

- a Why do chemists include commercial drugs on the same plate as the samples from the police?
 - b Does the confiscated sample contain any of the commercial drugs? Justify your answer using the data provided.
- 3 Describe the basic structure of a gas chromatograph.
 - 4 How are compounds identified in GC and HPLC?

Understanding

- 5 Explain how a gas chromatogram is produced.
- 6 Figure C6.36 shows the gas chromatogram for a sample of common pesticides. Predict which of compounds 1–5 is the smallest? Explain your reasoning.

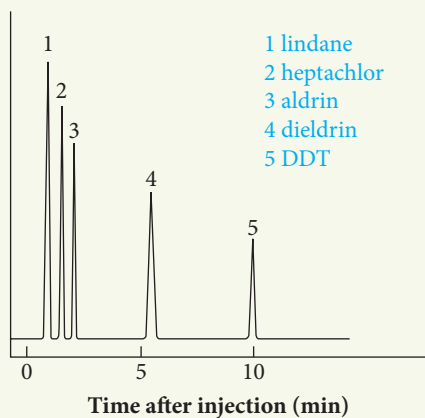
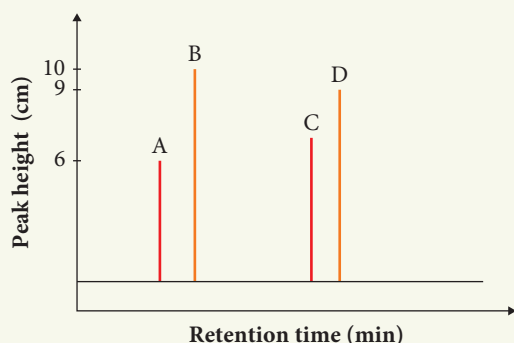


Figure C6.36 ◀

Gas chromatograms for some common pesticides

Applying

- 7 A mixture of pesticides was run through a gas chromatograph. The trace showed a very large peak at 4 minutes after injecting the sample, and smaller peaks between 1 and 3 minutes. What do you conclude about the composition of the sample?
- 8 Honeys vary in their composition. Honey consists generally of the disaccharides maltose and sucrose, and the monosaccharides glucose and fructose. HPLC is commonly used to analyse honey and to find out which sugars are present and in what percentage. The chromatogram in Figure C6.37 was obtained when a sample of Australian honey was tested.



◀ Figure C6.37

- Which molecule spent most time in the stationary phase? Explain how you determined this.
 - What is the percentage of C in the mixture?
 - Why was HPLC used in preference to GC in this analysis?
 - It is suspected that peak C is fructose. How can this be verified?
- 9 Table C6.5 gives the retention times and peak areas for a series of ethanol standards and a sample of champagne when they were run through a gas chromatograph.

Table C6.5

% Ethanol (w/v)	Relative peak area	R_t
8.4	4 343 500	0.90
9.8	5 820 300	0.89
10.6	6 858 600	0.90
12.2	8 117 900	0.89
Champagne sample	5 324 750	0.90

- Construct a calibration curve for the ethanol and use it to determine the percentage of ethanol in the champagne.
- If the density of ethanol is 0.785 g mL^{-1} , express your answer as % v/v.
- Explain why you need a calibration graph and not just peak areas.
- Explain why this calibration curve would not be suitable to find the alcohol content of a low alcohol wine that typically has a value of around 7% v/v.

10 A student's notebook was messy and it was hard to read the practical instructions on how to separate the individual dye components of food colouring. Not only was the order strange, but there were errors. The instructions were as follows.

- I** Using a gel pen, draw a reference line 2 cm from the bottom of a strip of chromatography paper.
- II** Put more drops of the food colouring over the first to make a concentrated spot that is at least 15 mm in diameter so that it has adequate contact with the paper and can be easily seen.
- III** Allow the drop to dry.
- IV** Carefully place a drop of food colouring on the line.
- V** Place solvent in the bottom of a tall jar.
- VI** Hang the strip of chromatography paper in the jar, making sure that the spot of colouring is completely covered by the solvent.
- VII** Allow the paper to remain in contact with the solvent until a solvent front has almost reached the top of the paper.
- VIII** Observe the separation of the coloured components of the food colouring. Note that the component that absorbs most strongly to the paper pulls itself furthest up the paper.
- IX** Calculate an R_f value for each spot on the chromatogram.
- X** Identify the components of the food colouring by comparing the R_f value with those of known dyes.
 - a** Identify the correct order for the instructions (I–X).
 - b** Identify three fundamental errors in the instructions and in each case explain how the error should be corrected.
 - c** Explain how R_f values are calculated.

Reflecting

11 What challenges and opportunities have arisen from the development of HPLC and GC methods in the detection of drugs and other banned substances in sport?

CHAPTER CHECKLIST

You should know:

- molecules can be represented by electron dot formulas and valence structures
- VSEPR theory can be used to help determine the structure of molecules and some ion complexes
- polarity is due to the presence of a permanent dipole in molecules and asymmetry in the molecule
- intermolecular forces of dispersion, dipole–dipole and hydrogen bonding are present in molecules
- dipole–dipole interactions occur when there is an unequal sharing of electron density between different elements
- hydrogen bonding is a special case of dipole–dipole interactions. It is due to the uneven sharing of electrons between hydrogen and a strongly electronegative element (N, O and F)
- dispersion forces exist between all molecules and non-metal atoms
- hydrogen bonding in water is responsible for the unique properties of water
- water is the universal solvent
- surfactants and detergents affect surface tension
- the interactions between molecules are the basis for practical applications such as the chromatography techniques TLC, GC and HPLC
- chromatography techniques (HPLC, GC and TLC) rely on absorption onto a stationary phase and desorption back into a mobile phase
- by running a standard under the same conditions, calculation of the R_f and R_i can identify the species present
- the use of calibration curves of the amount under the peak at a particular R_f can be used to determine the amount present in a sample.

You should be able to:

- predict whether a molecule will have dipole–dipole, hydrogen bonding or dispersion forces between the molecules
- link the intermolecular bonding present to properties such as solubility
- draw electron dot formulas and valence structures of compounds and simple molecules
- determine the shape of the compounds and simple molecules using the VSEPR model
- determine whether a solution is saturated, unsaturated or supersaturated by using a solubility curve
- determine the concentration of solutes in the units ppm, % w/w, % w/v and % v/v
- interpret chromatograms, including the use of R_f and R_i for particular components
- use calibration curves to determine the amount of a component in a chromatograph
- know which instrument (HPLC or GC) to choose for a particular application.

CHAPTER GLOSSARY

adhesive force the electrostatic force of attraction between unlike particles as displayed between water molecules and glass

adsorb to be attracted to the surface of a material; the opposite of desorb

analyte the substance being analysed

anhydrous contains no water

anionic negatively charged; for example, Cl^-

bonding pair a pair of electrons shared by two atoms to form a covalent bond

capillary action the process in which a liquid is drawn up a narrow tube

cationic positively charged; for example, Na^+

cohesive force the electrostatic force of attraction between like particles, which causes water molecules to stick to each other

colloid a mixture in which tiny clusters of particles are dispersed through another substance; colloids do not settle out due to gravity and the particles are too small to be filtered

desorb the action of the substance moving from the stationary phase into the mobile phase

dipole a permanent build-up of negative charge at one end, and positive charge at another end, of a covalent bond or molecule

dipole-dipole force the attraction between molecules with permanent dipoles

dissociated when ionic salts in a solvent dissolve, ions of opposite charge no longer associate with each other and move freely

dispersion force the weak attractive force between atoms and molecules caused by an instantaneous temporary change in dipole moment, arising from the movement of orbiting electrons

electrolytic solution a solution that can conduct electricity due to the presence of ions

electromagnetic attraction the attraction of oppositely charged particles

electronegativity the ability of an atom to attract electrons

elute movement of a mobile phase through a column

emulsion a mixture of two substances that would not normally mix

hard water water that contains high levels of Ca^{2+} and Mg^{2+} ions, which interferes with the action of soaps

heteroatomic a molecule containing more than one type of element

hydrogen bond the intermolecular attraction between hydrogen in $-\text{OH}$, $-\text{NH}$ or $-\text{FH}$ and the lone pairs on nitrogen, oxygen or fluorine on an adjacent molecule

hydrophilic 'water-loving', a particle with polar regions that bond to water

hydrophobic 'water-hating', a particle with mostly non-polar regions that do not bond with water

immiscible liquids that do not mix

intermolecular forces forces between individual particles (molecules, ions, atoms) of a substance

ion-dipole attraction between the charge on one ion and a dipole in another molecule

ligand an ion or molecule attached to a metal atom (or larger molecule) by dipole bonding

lone pair a pair of valence electrons that are not involved in bonding

meniscus the shape of the surface of a liquid near the edge of a container

micelle small spherical cluster of surfactant molecules suspended in a liquid

mobile phase the phase that carries solutes through the stationary phase

non-ionic detergent a surfactant with a non-charged hydrophilic part

origin in chromatography, the line that samples are placed on; the distance the solvent moves is measured from here

Pauli exclusion principle an orbital can contain 0, 1 or 2 electrons

polar a molecule with a separation of charge; one end positive, another region negative

polar covalent bond a covalent bond that has a separation of charge

R_f retardation factor; the distance a solute moves compared to the distance the solvent has moved

R_t retention time; the time it takes for a solute to elute from the column in GC and HPLC

saponification the production of soap by mixing fats with strong alkali

saturated solution a solution that has the most solute possible at a particular temperature

solubility the amount of solute that can dissolve in 100 g of water (solvent)

solubility curve a graph showing how solubility changes with temperature

stationary phase the substance that is fixed in place during chromatography, to which the solute adsorbs

structural formula a chemical formula that shows how atoms are linked

supersaturated solution an unstable solution that has more solute than possible at that temperature; it is formed from dissolving solute into the solution at a higher temperature and then allowing it to cool. The extra solute is still in solution but can readily be crystallised out

surface tension the force that arises from the attraction of the surface molecules to the bulk of the material

surfactant a chemical that lowers the surface tension of a liquid

suspension a cloudy, heterogeneous mixture containing solid particles that will eventually settle out

unsaturated solution a solution into which more solute can dissolve

valence structure a diagram of the arrangement of atoms around a central atom

vapour pressure pressure exerted down by a vapour onto its liquid form

CHAPTER REVIEW QUESTIONS

Remembering

- 1 What are the two types of bonding in liquid propane?
- 2 Draw a diagram showing the bonding in ice.
- 3 A mixture of two substances is separated by paper chromatography. The substance with the greater R_f value:
A moves further and is more strongly adsorbed on the paper.
B moves further and is less strongly adsorbed on the paper.
C moves less and is more strongly adsorbed on the paper.
D moves less and is less strongly adsorbed on the paper.
- 4 Draw the electron dot formula for the following and indicate the number of bonding and non-bonding pairs of electrons.
a NO_2^-
b CO
c CH_4
- 5 Draw valence structures for the following molecules and indicate whether they are polar or non-polar.
a CO_2
b F_2O
c CH_4
d PF_3
- 6 Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) and glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) can dissolve in water. Write the chemical equations that represent this.
- 7 Describe what happens when an ionic substance such as potassium iodide dissolves in water.
- 8 Formamide is a liquid with a very high surface tension, which wets glass. If the end of a thin glass capillary is dipped into a beaker of formamide, what would be observed?

Understanding

- 9 In a mixture of alcohols, each component can be identified by HPLC. The alcohol with the longest retention time will be the one that:
A is the most soluble in the mobile phase.
B has the smallest molar mass.
C will emerge first from the column.
D adsorbs most strongly on the stationary phase.
- 10 Explain the existence of the property of the surface tension of a liquid in terms of the interactions between the molecules.
- 11 You are asked to perform a simple experiment to determine which has the greater surface tension, water or diesel.
a Describe the method you would use.
b Describe the results you would expect to observe.
c Provide an explanation for these results.
- 12 Explain why a needle can float on water in a Petri dish until a drop of detergent is added.
- 13 Calculate the total volume of alcohol in a small bottle (375 mL) of wine when the alcohol content is 8.5% v/v.
- 14 Hydrogen peroxide is a strong bleach used for hair and whitening teeth. Would you expect it to be soluble in water? Explain with the use diagrams the forces that might operate between the molecules of peroxide and water.

- 15 Use Figure C6.25 on page 280 to determine the mass of potassium nitrate that will dissolve in:
- a 100 g of water to form a saturated solution at 20°C.
 - b 50 g of water to form a supersaturated solution at 30°C.
 - c 80 g of water to form a saturated solution at 10°C.
- 16 Serotonin ($\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}$) is a compound that conducts nerve impulses in the brain and muscles. A sample of spinal fluid was found to contain a serotonin concentration of 1.5 ng L^{-1} . How much serotonin is there in 1 mL of spinal fluid?

Applying

- 17 Which one of the following is a property of a 'water in oil' emulsion?
- A It feels greasy to the touch.
 - B It mixes readily with water.
 - C Its electrical conductivity is higher than that of an 'oil in water' emulsion.
 - D It can be coloured by the addition of a water-soluble dye.
- 18 Describe the trend in boiling points of the noble gases and halogens in Table C6.2. How does this relate to intermolecular bonding?
- 19 Figure C6.25 on page 280 shows the solubility curve of potassium chloride, KCl.
- a Use diagrams to model how KCl would dissolve in water.
 - b Write a balanced equation for the dissolution of KCl in water.
 - c How much KCl would just dissolve in 50 mL of water at 80°C?
 - d If the solution in part c was cooled to 10°C, would it be saturated, unsaturated or supersaturated? Justify your answer.
 - e What temperature would you need to heat 100 g of water so that 40 g of KCl would dissolve completely? If you cooled this to 10°C, how much KCl would precipitate out of solution?
- 20 A commercial liquid medication is known to contain 400 mg of magnesium hydroxide per 10.0 mL of the medication. Express this as % w/v and ppm. (Assume that 1 mL of medication = 1 gram.)
- 21 Explain why different solutes travel through a gas chromatograph at different rates.

Analysing

- 22 Steroids do not circulate in the blood system by themselves. Infer the polarity of steroid hormones. Dopamine can travel through the blood system easily. What does this indicate about the polarity of dopamine? Which of these is more likely to be injected directly into the veins?
- 23 Compare and contrast ammonia (NH_3) and methane (CH_4). Highlight the bonding so that you can explain to another student how to determine which one is soluble in water.
- 24 Reorganise the data in Table C6.1 on page 262 to clearly show the relationship between the generic formula and the lone pair of electrons.

Reflecting

- 25 Theories such as VSEPR have enabled scientists to develop a greater understanding of molecular structures. Representations have changed from valence structures to three-dimensional representations of electrons as charged clouds. Explain how theories and models generally develop over time, using this specific theory.